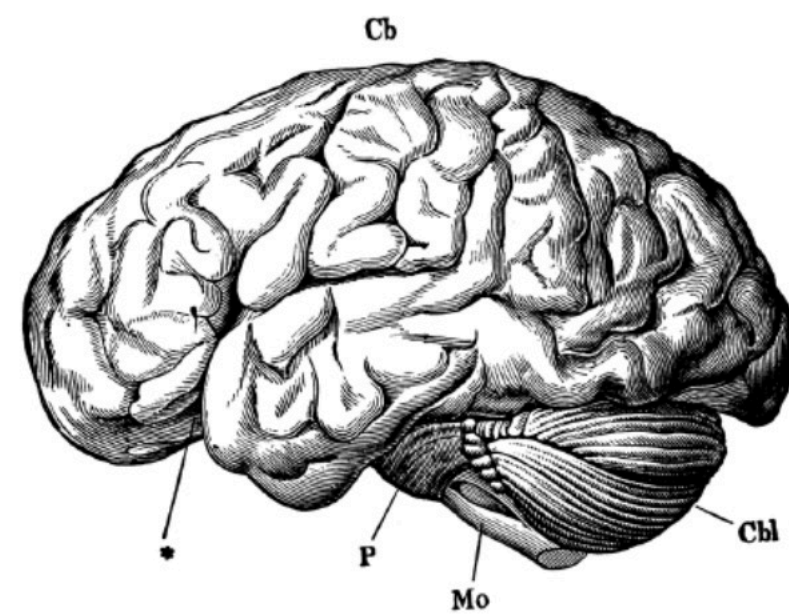


Directed Acyclic Graphs

DAEN 427 / ISEN 427 / ISEN 627 Fall 2026

Alexander Abuabara



**Human
language**

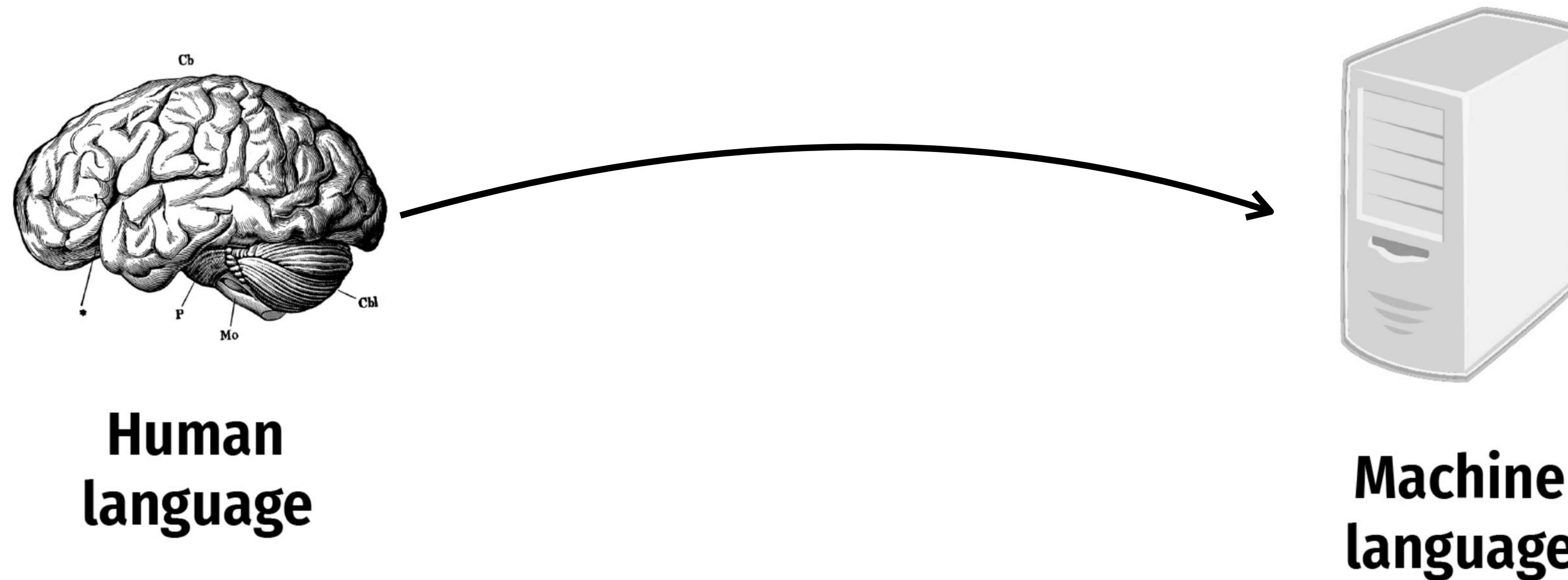


**Machine
language**

Speak and do causation

Design rigorous evaluations

**Change the world
with data**



In lists of key employability skills, **problem solving** usually appears at the top (makes sense). What is the difference between problem-solving and problem-structuring?

The key to answering this question is not in a protracted wrangle about the differences but in examining the nature of problems problem-structuring involves **defining, organizing, and clarifying** a problem before attempting to solve it, ensuring a more effective resolution!

Causal Revolution

Indicators

Causal Models

Paths, Doors, and Adjustments

Decision Models

Indicators

Godwin's Law

Godwin's law

From Wikipedia, the free encyclopedia

Godwin's law (or **Godwin's rule of Hitler analogies**)^{[1][2]} is an Internet **adage** asserting that "as an online discussion grows longer, the probability of a **comparison involving Nazis or Hitler approaches 1**".^{[2][3]} That is, if an online discussion (regardless of topic or scope) goes on long enough, sooner or later someone will compare someone or something to **Adolf Hitler** or his deeds, the point at which effectively the discussion or thread often ends.

Promulgated by the American attorney and author **Mike Godwin** in 1990,^[2] Godwin's law originally

Godwin's Law for statistics

SCIENCE

The Internet Blowhard's Favorite Phrase

Why do people love to say that correlation does not imply causation?

By DANIEL ENGBER

OCT 02, 2012 • 8:33 AM

Not everyone found the news believable. “Facepalm. Correlation doesn't imply causation,” wrote one unhappy Internet user. “That’s pretty much how I read this too... correlation is NOT causation,” agreed a Huffington Post superuser, seemingly distraught. “I was surprised not to find a discussion of correlation vs. causation,” cried someone at Hacker News. “Correlation does not mean causation,” a reader moaned at Slashdot. “There are so many variables here that it isn’t funny.”

Correlation does not
imply causation

Except when it does

Even if it doesn't,
this phrase is useless
and kills discussion

Correlation vs. Causation

How do we figure out correlation?

How do we figure out causation?

Correlation vs. Causation

How do we figure out correlation?

Math and statistics

How do we figure out causation?

Philosophy.
No math.

Causal relationships?

Lighting fireworks causes noise

Rooster crows cause the sunrise

Getting an MPA increases your earnings

Colds go away a few days
after you take vitamin C

Causation

**Causation =
Correlation + time order + nonspuriousness**

nonspuriousness means that a relationship observed between two variables is not due to the influence of a third

How do you know if you have it right?

You need a philosophical model

That's what this class is for!



John B. Holbein @JohnHolbein1 · Apr 7

Causality isn't achieved; it's approached.



3



1



8



[Show this thread](#)



John B. Holbein @JohnHolbein1 · Apr 7

Causality isn't binary; it's a continuum.



1



5



13



[Show this thread](#)

Essencial parts

Inputs, activities, and outputs

Generally directly
measurable

of citations mailed,
% increase in grades, etc.

Outcomes

Harder to measure directly

Loftier and more abstract

Commitment to school,
reduced risk factors

Indicators

Y

Outcome variable

Response variable

Dependent variable

**Thing you want to
explain or predict**

X

Explanatory variable

Predictor variable

Independent variable

**Thing you use to
explain or predict Y**

Y "listens to" X

"A variable X is a cause of a variable Y if Y in any way relies on X for its value. ... X is a cause of Y if Y listens to X and decides its value in response to what it hears"

(Pearl, Glymour, and Jewell 2016, 5–6)

Y doesn't necessarily listen only to X

**A light switch causes a light to go on,
but not if the bulb is burned out (no Y despite X),
or if the light was already on (Y without X)**

Two purposes of modeling

Prediction

Forecast the future

Focus is on Y

Netflix trying to
guess your next show

Predicting who will enroll in SNAP

Explanation

Explain effect of X on Y

Focus is on X

Netflix looking at the effect of the
time of day on show selection

Measuring the effect of
SNAP on poverty reduction

**How do you measure
abstract outcomes?**

Move up the ladder of abstraction.

Ladder of abstraction for witches



Top rung: Broad, conceptual, hard to measure directly.
Bottom rung: Specific, tangible, observable.

Ladder of abstraction for witches

Mammal

Enmagicked

Trolls, elves, gods/goddesses

Female

Arwen, Winky, Athena

Human

Young

Salem witch trials

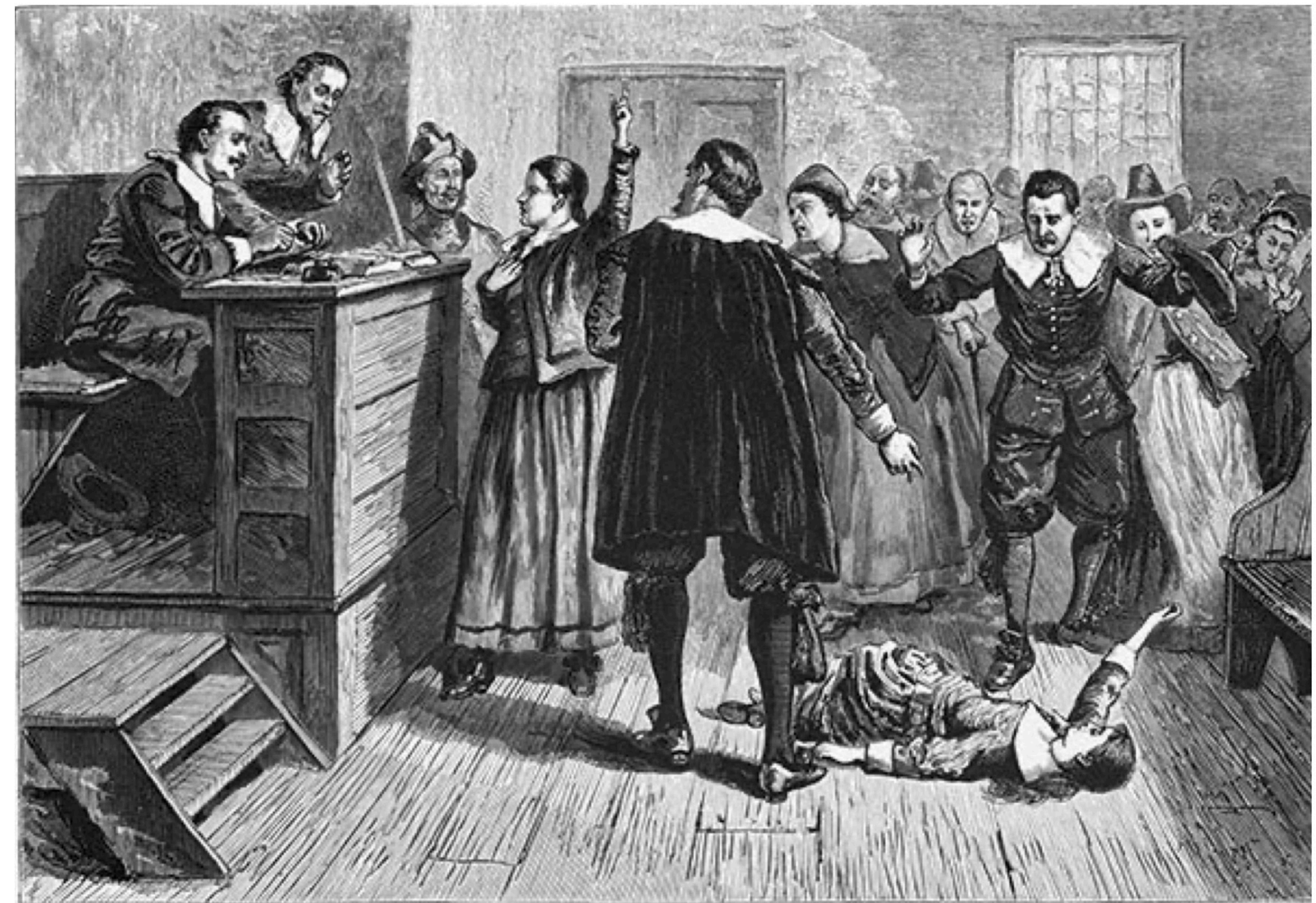
Student

Hermione Granger,
Sabrina Spellman

Old

Elphaba, Halloween
decorations

Connection to theory





"*My Wife and My Mother-in-Law*" is a famous optical illusion that depicts both an old woman looking off to the left and a young woman facing away, looking over her right shoulder. (The old woman's nose is the young woman's chin.)
(Image credit: Public Domain)

Source: <https://www.livescience.com/63645-optical-illusion-young-old-woman.html>

Causal models

Types of data

Experimental

**You have control over
which units get treatment**

Observational

**You don't have control
over which units get
treatment**

**Which kind lets you
prove causation?**

Causation with observational data

Can you prove causation with observational data?

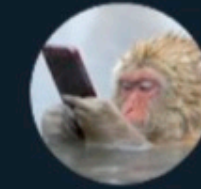
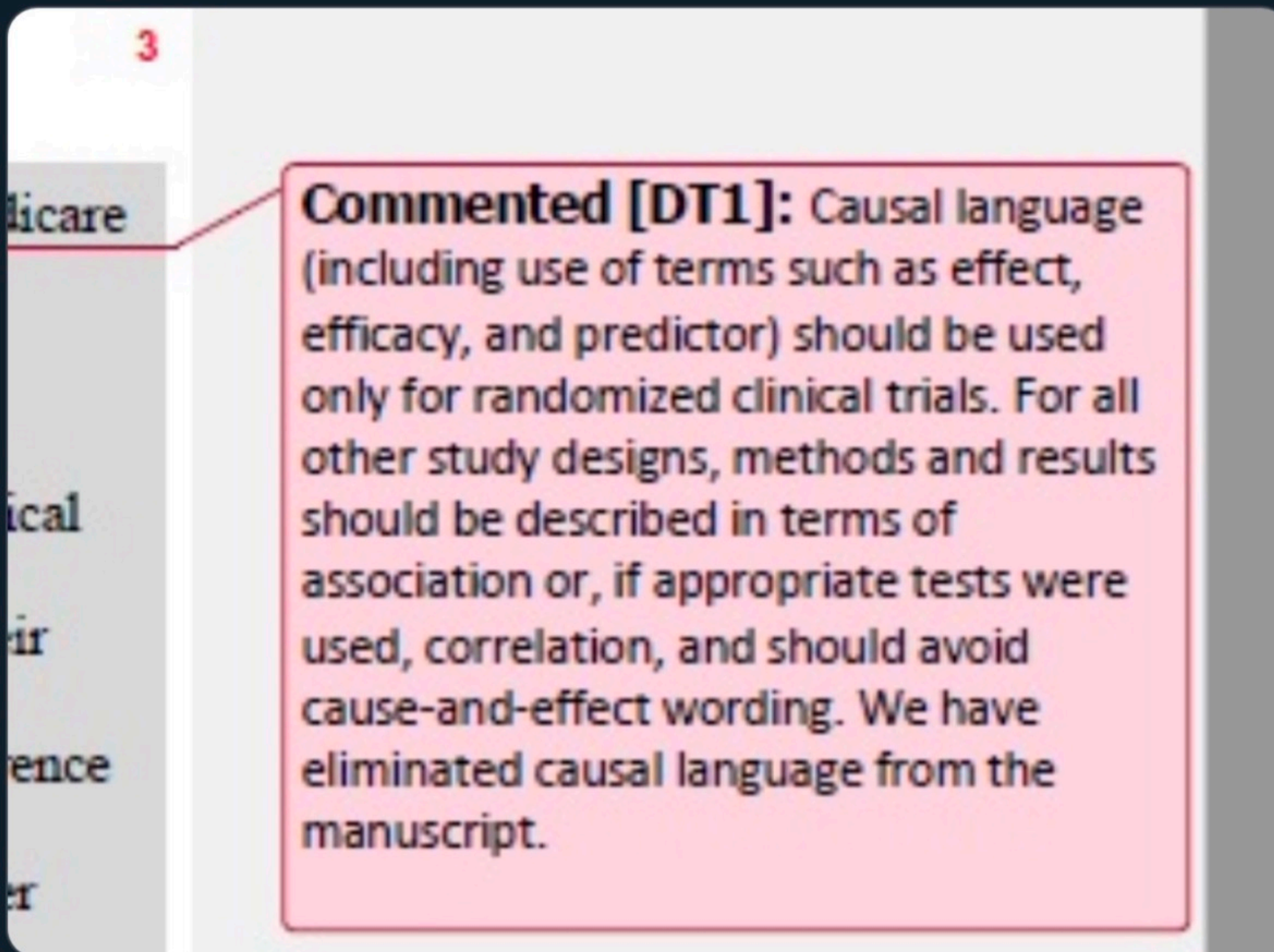
Why is it so controversial to use observational data?



Laura Hatfield
@laura_tastic

Wow: this comment from fresh page proofs.

Guess all of us researching causal inference in observational data need to find new jobs?



Seva
@SevaUT

normal person: this rain is making us wet

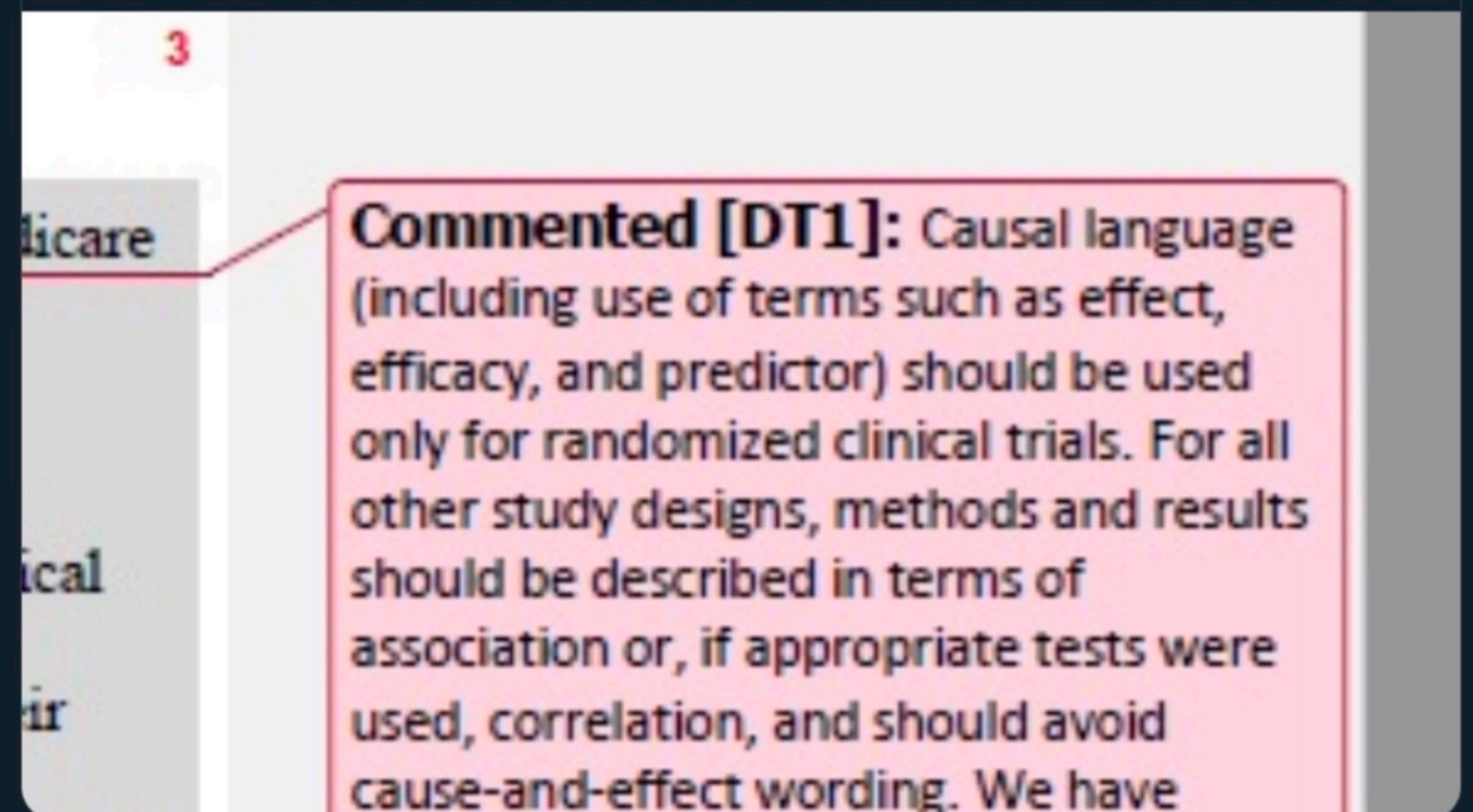
me, RCT genius: whoa there! First, take twenty walks and randomly apply the rain treatment



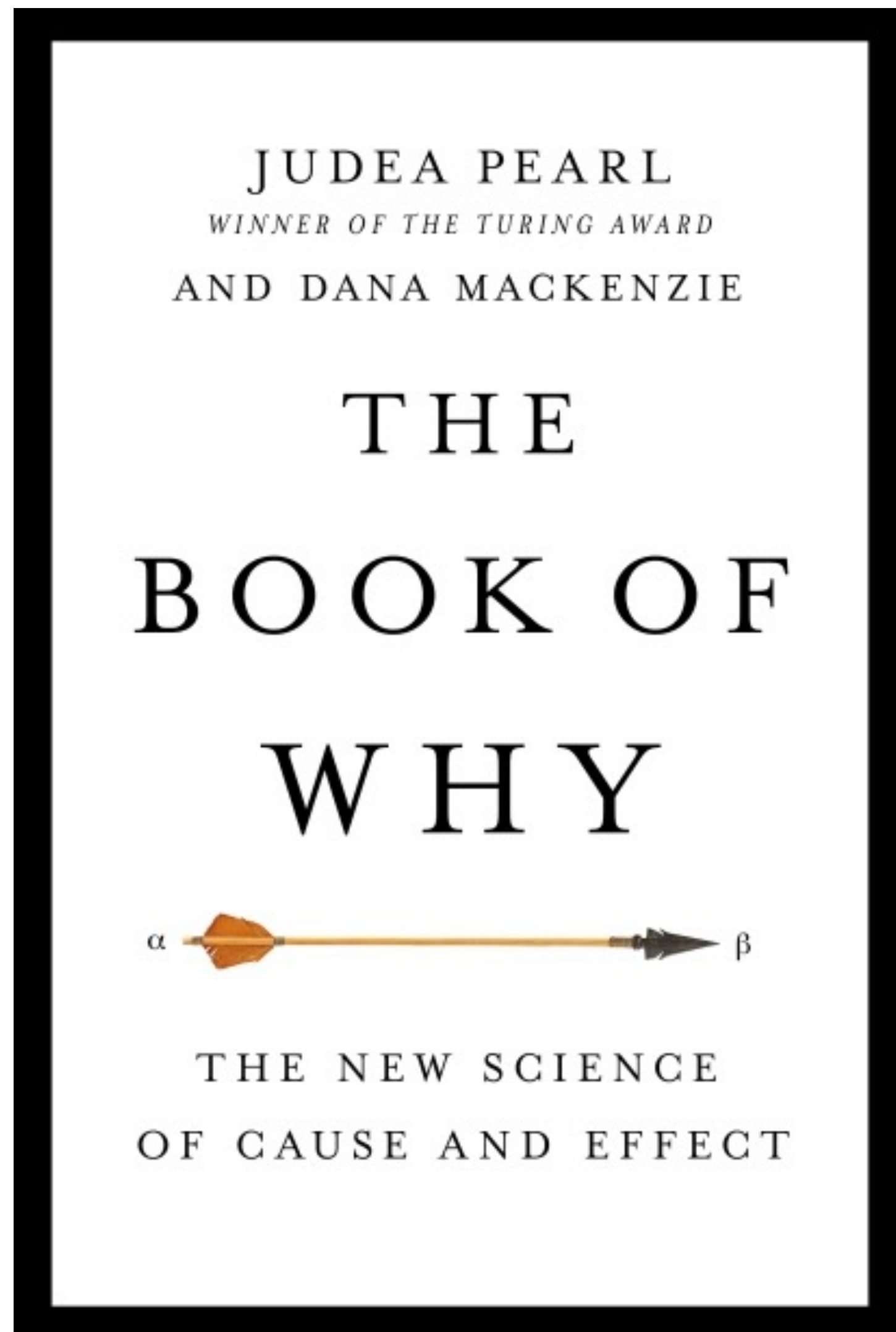
Laura Hatfield @laura_tastic · Jan 16

Wow: this comment from fresh page proofs.

Guess all of us researching causal inference in observational data need to find new jobs?



The causal revolution



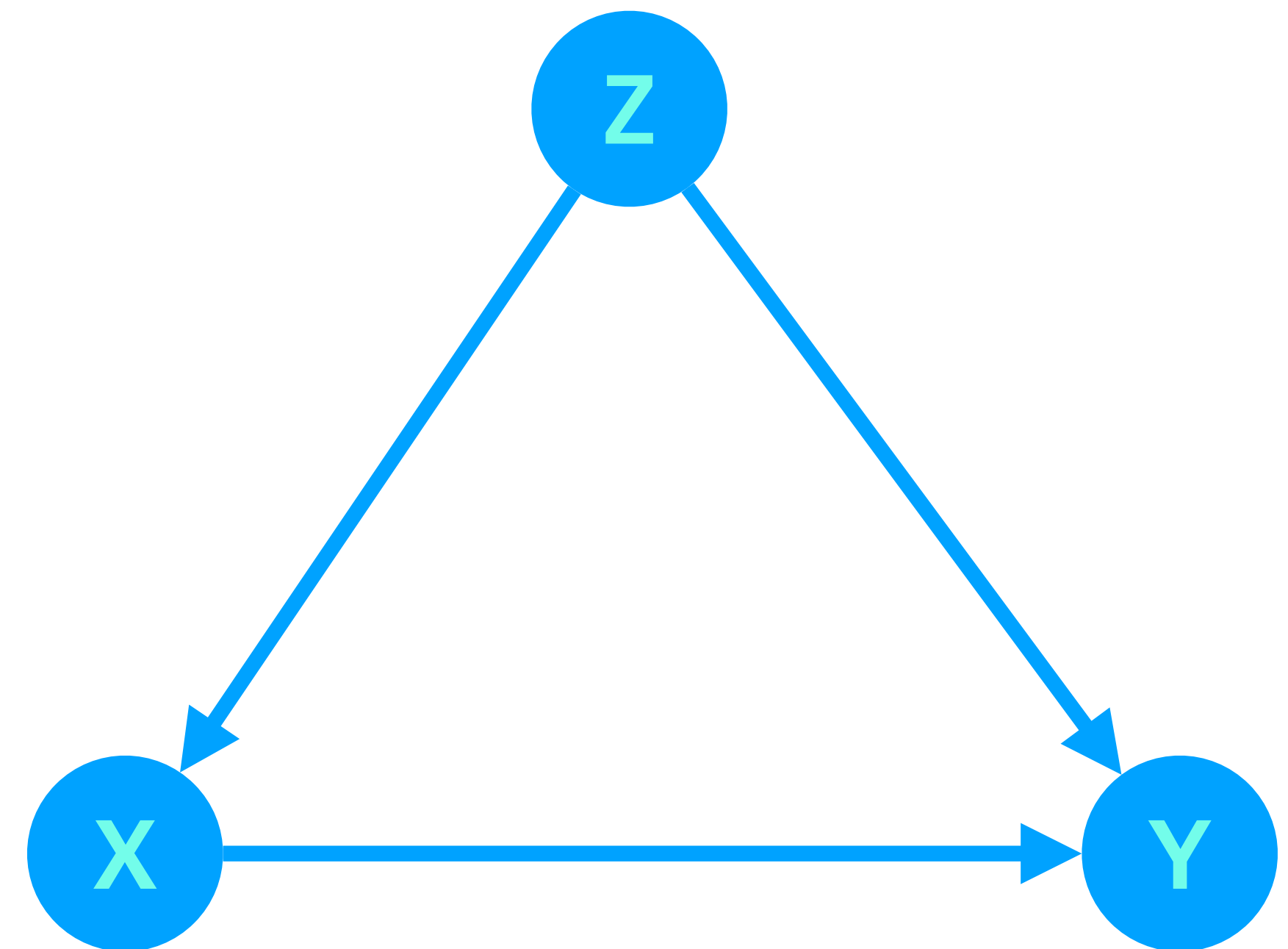
Causal diagrams

Directed acyclic graphs (DAGs)

Directed: Each node has an arrow that points to another node

Acyclic: You can't cycle back to a node (and arrows only have one direction, causal effects cannot be bidirectional)

Graph: It's... um... a graph



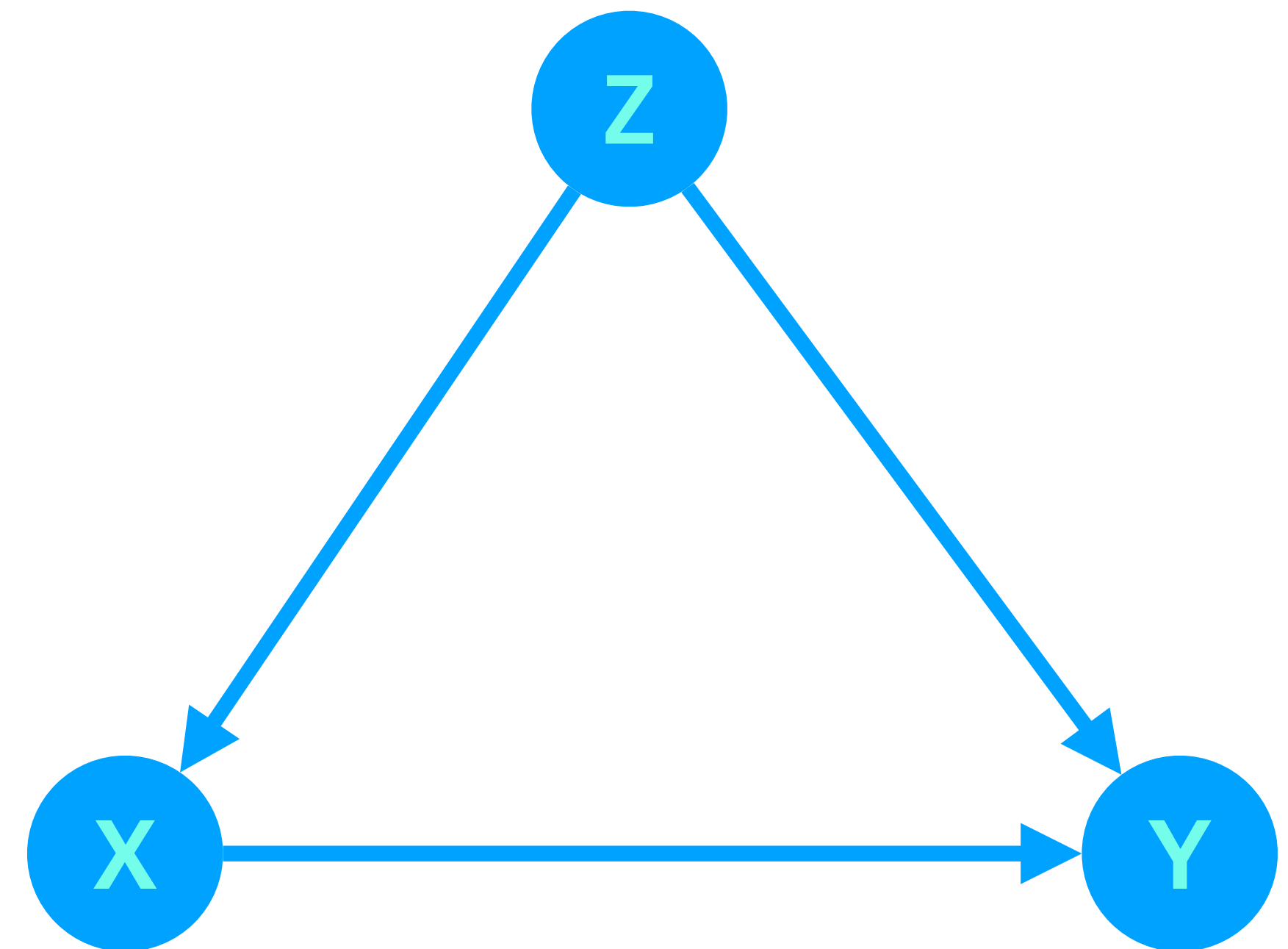
Causal diagrams

Directed acyclic graphs (DAGs)

Graphical model of the process that generates the data

Maps your philosophical model

Fancy math ("*do*-calculus") tells you what to control for to isolate and identify causation



Causal diagrams

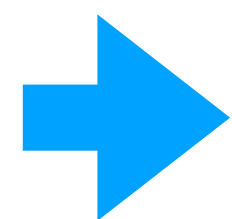
Directed acyclic graphs (DAGs)

Computer science: data structure

Markov models: visualization

Explicitly identify assumptions

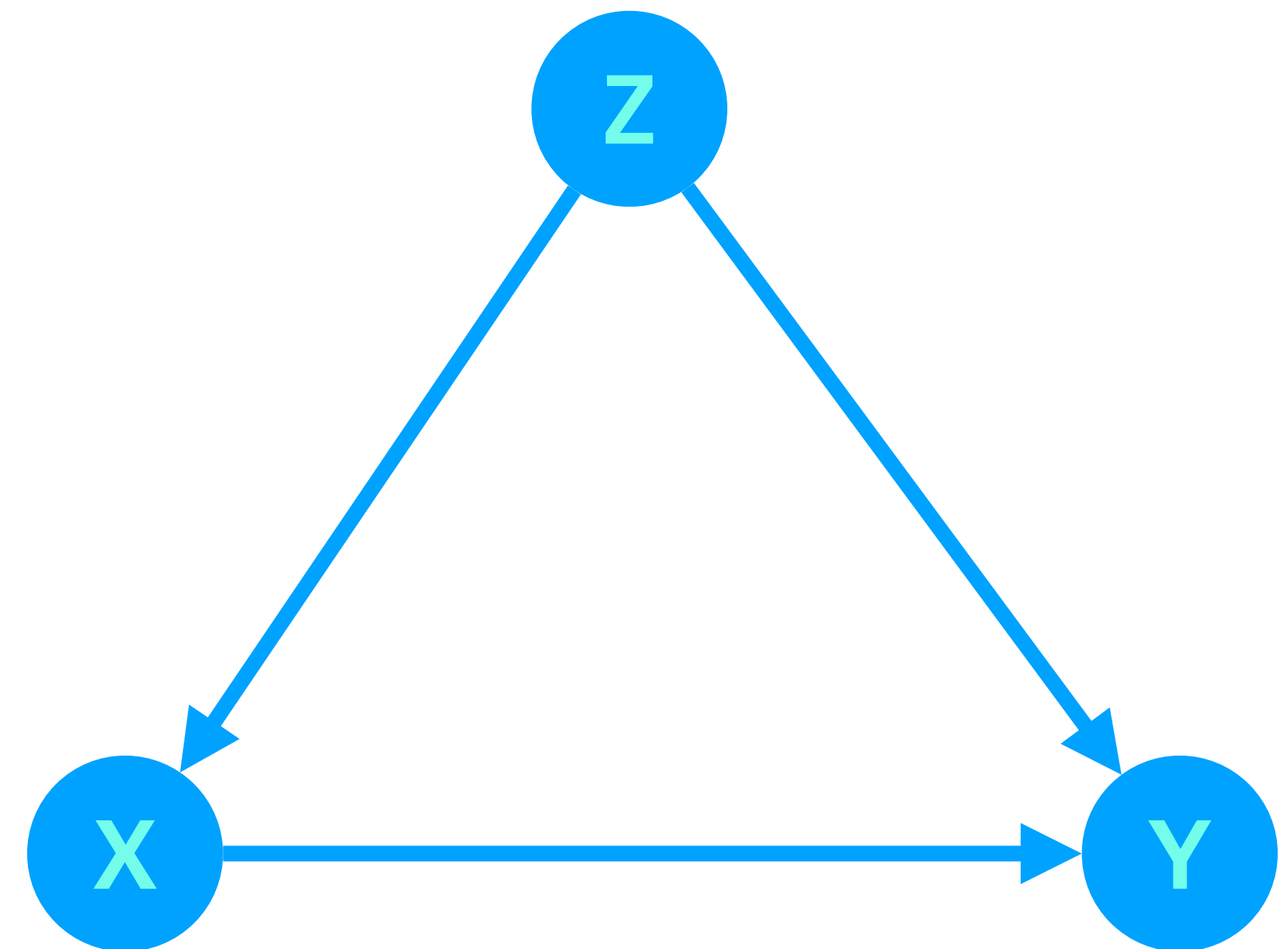
Illustrate sources of bias



Reduce bias

Separate individual effects

Ascertain appropriate covariates for statistical analysis

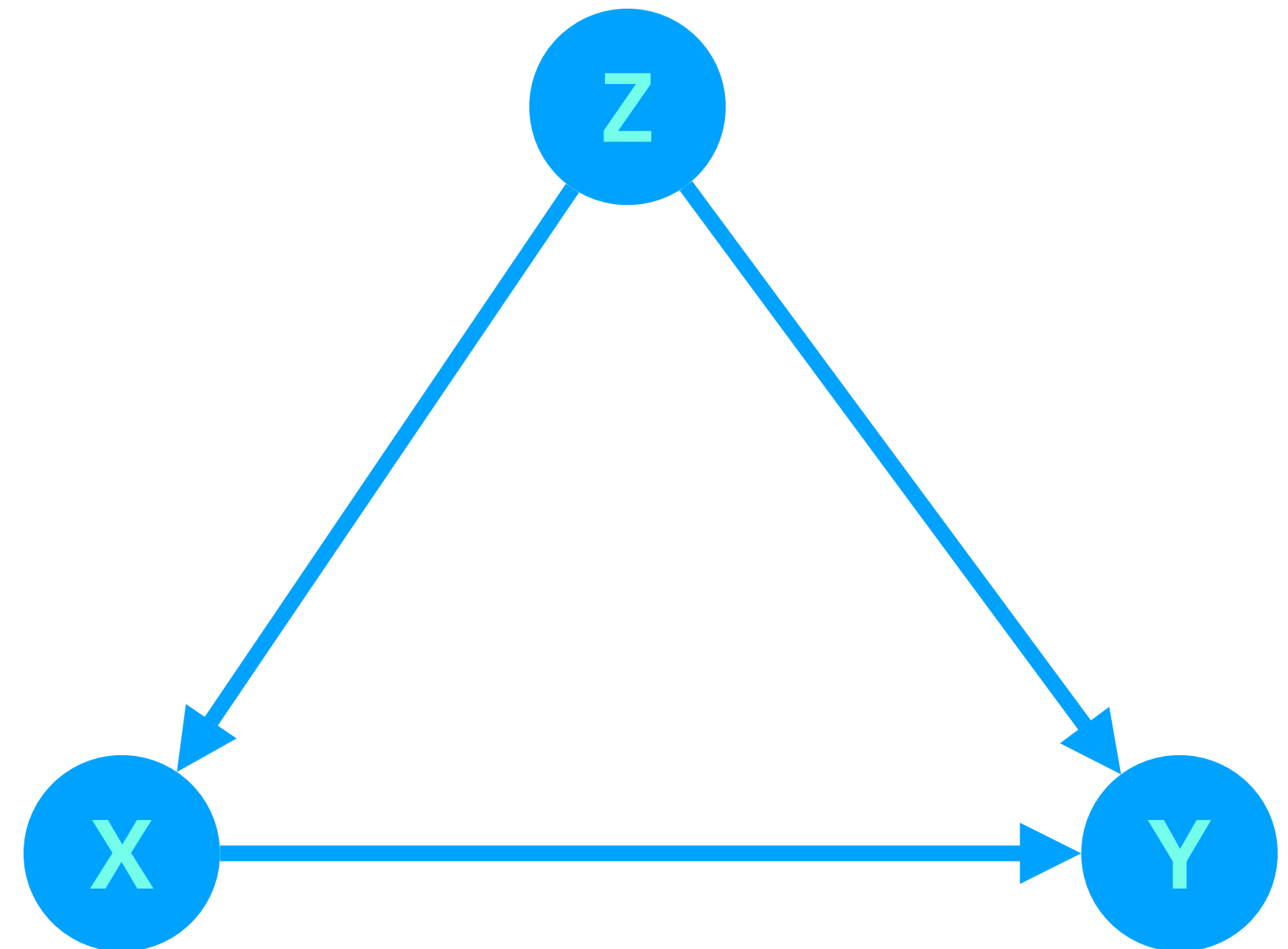


Causal diagrams

Directed acyclic graphs (DAGs)

Causal graphs are
based on assumptions

But so are analytic
models



Acyclicalness

What if there's something that really is cyclical?

Wealth $\rightarrow$ Power $\rightarrow$ Wealth



This isn't acyclic!
Wealth $\leftrightarrow$ Power

Split the node into different time periods

Wealth _{$t-1$} $\rightarrow$ Power _{t} $\rightarrow$ Wealth _{t}

How to draw a DAG

What is the causal effect of an additional year of education on earnings?

Step 1: List variables

Step 2: Simplify

Step 3: Connect arrows

Step 4: Use logic and math to determine which nodes and arrows to measure

1. List variables

Education (treatment) → Earnings (outcome)

Location

Ability

Demographics

Socioeconomic status

Year of birth

Compulsory schooling laws

Job connections

2. Simplify

Education (treatment) → Earnings (outcome)

Location

Ability

Demographics

Socioeconomic status

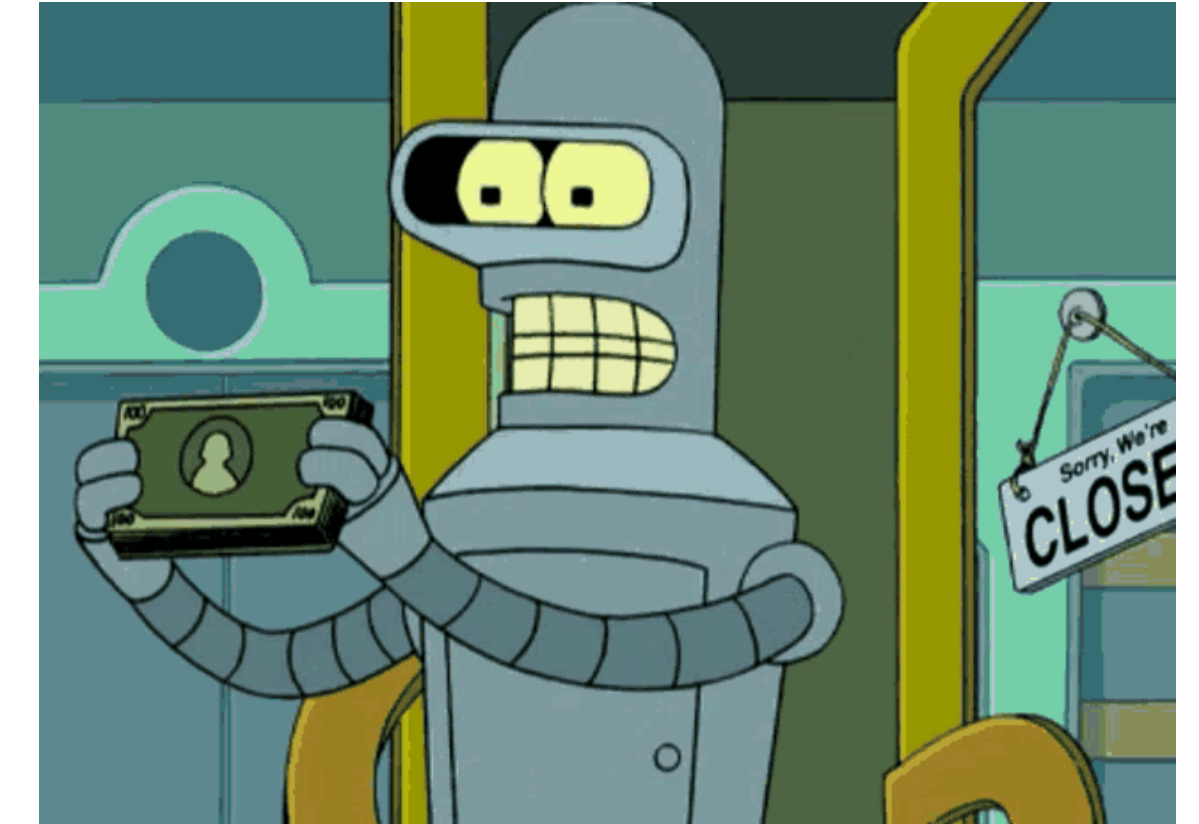
Year of birth

Compulsory schooling laws

Job connections

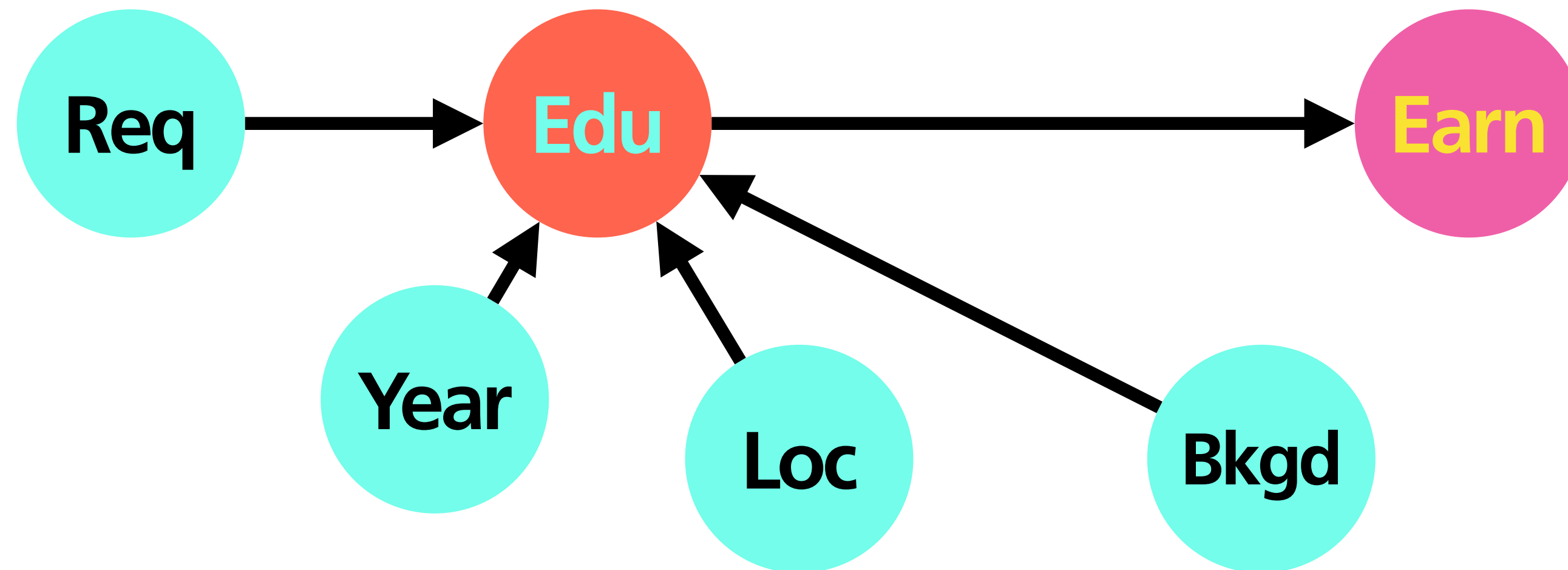
Background

3. Draw arrows



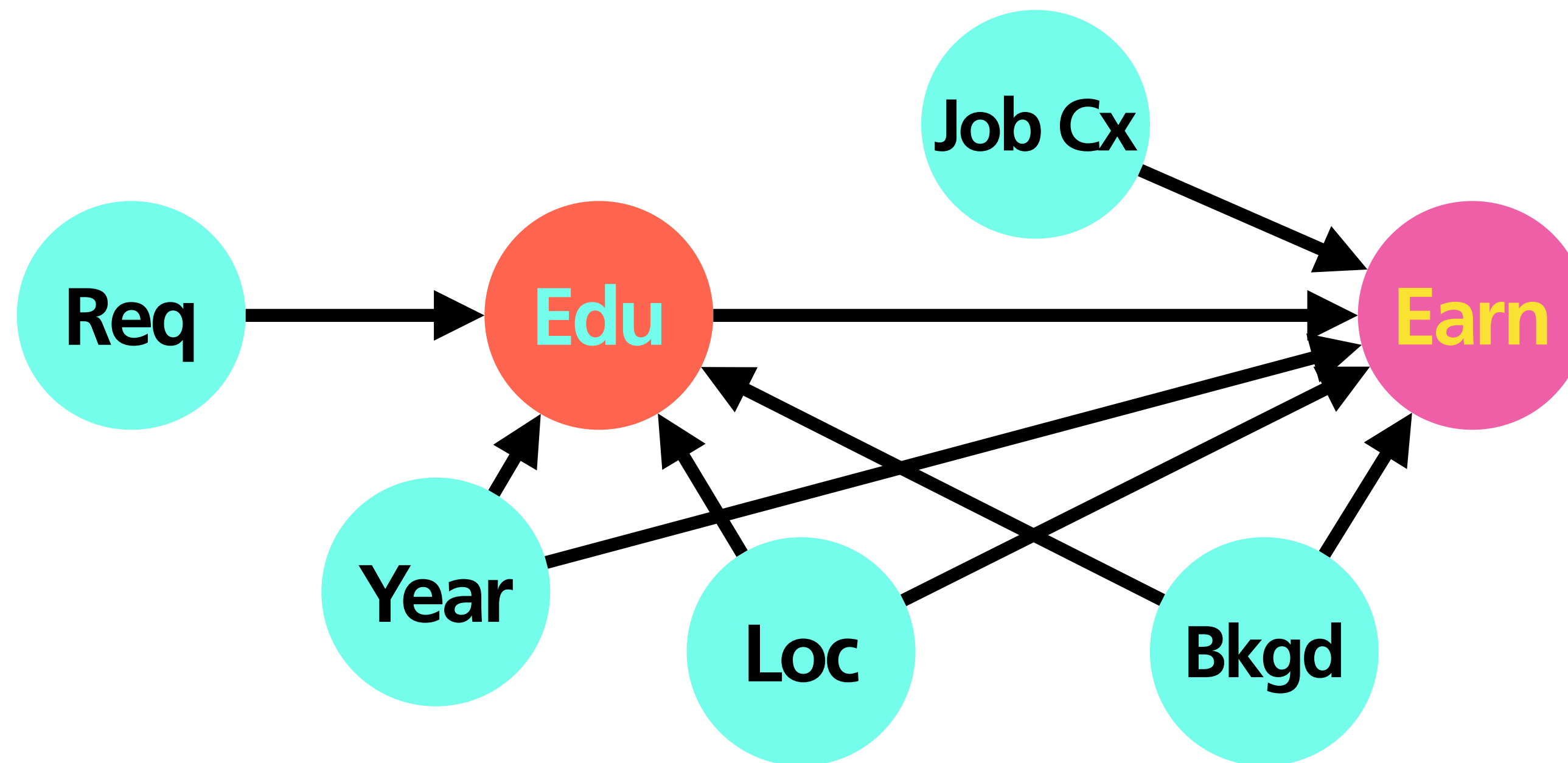
Education causes earnings

3. Draw arrows



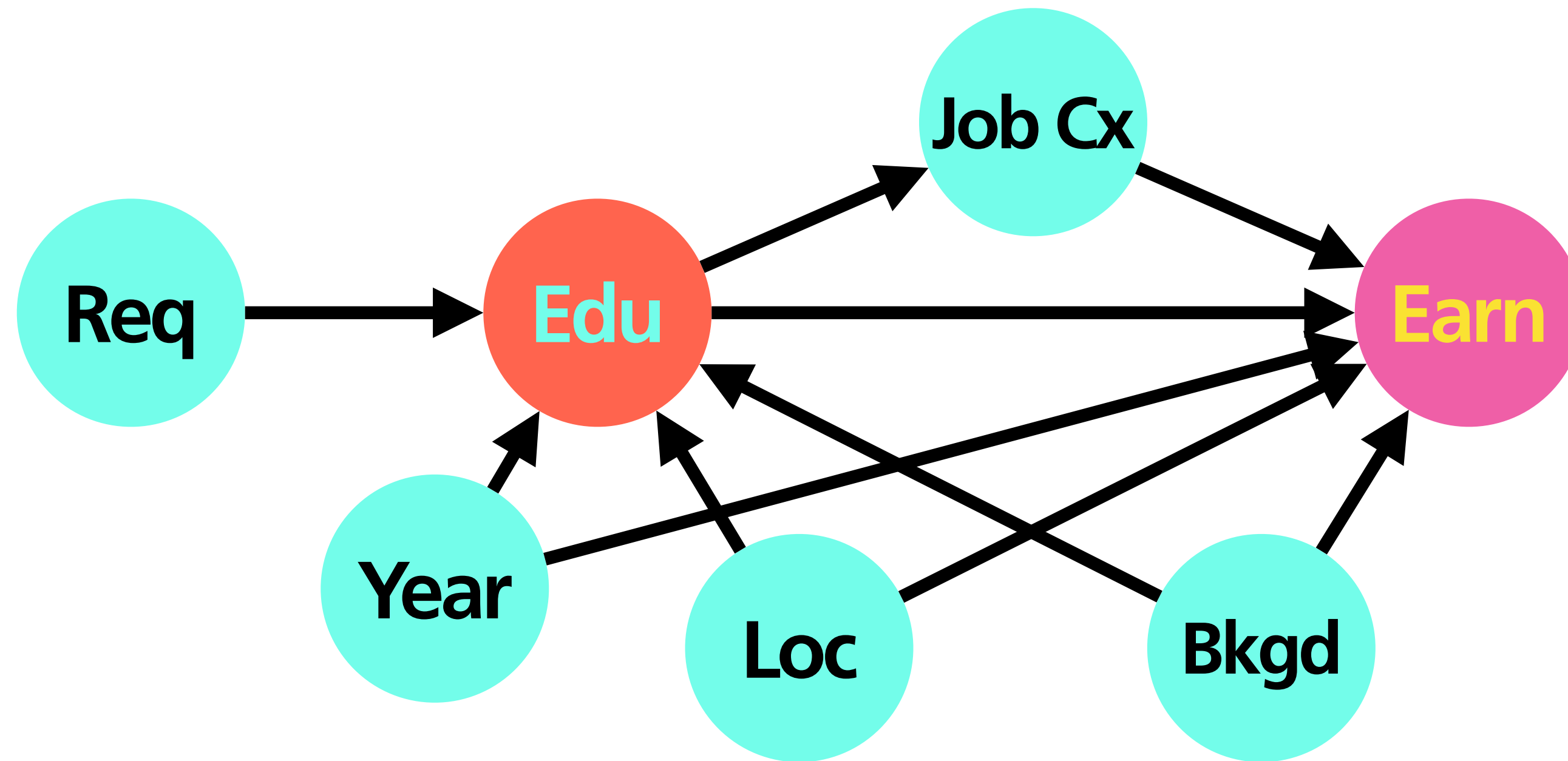
Background, year of birth, location, and school requirements all cause education

3. Draw arrows



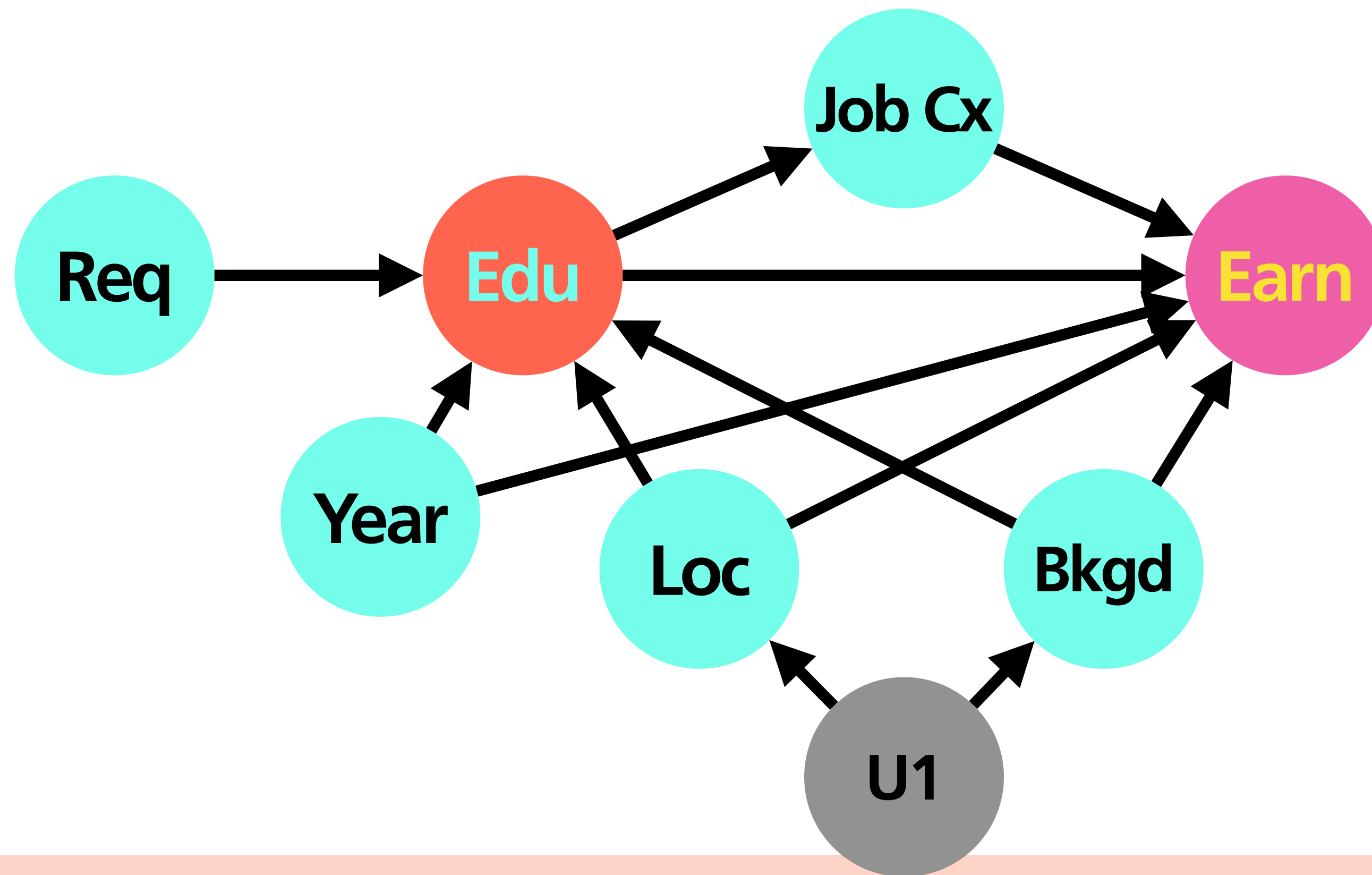
Background, year of birth, and location all
cause earnings too

3. Draw arrows



Education causes
job connections too

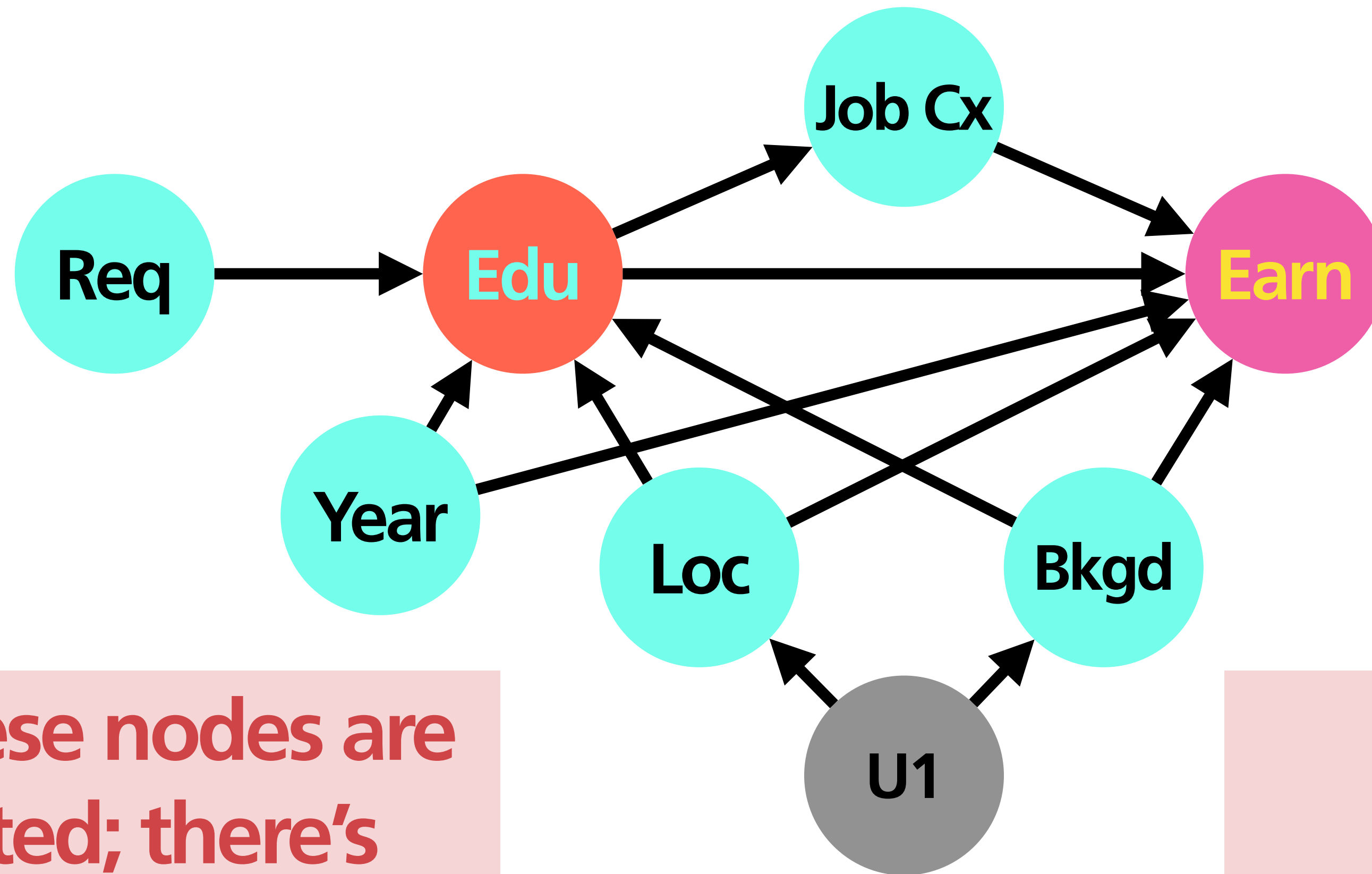
3. Draw arrows



Location and background are probably related, but neither causes the other. Something unobservable (**U1**) does that.

**Paths, doors,
and adjustments**

Causal identification



All these nodes are related; there's correlation between them all

We care about $\text{Edu} \rightarrow \text{Earn}$, but what do we do about all the other nodes?

Causal identification

A causal effect is *identified* if the association between treatment and outcome is properly stripped and isolated

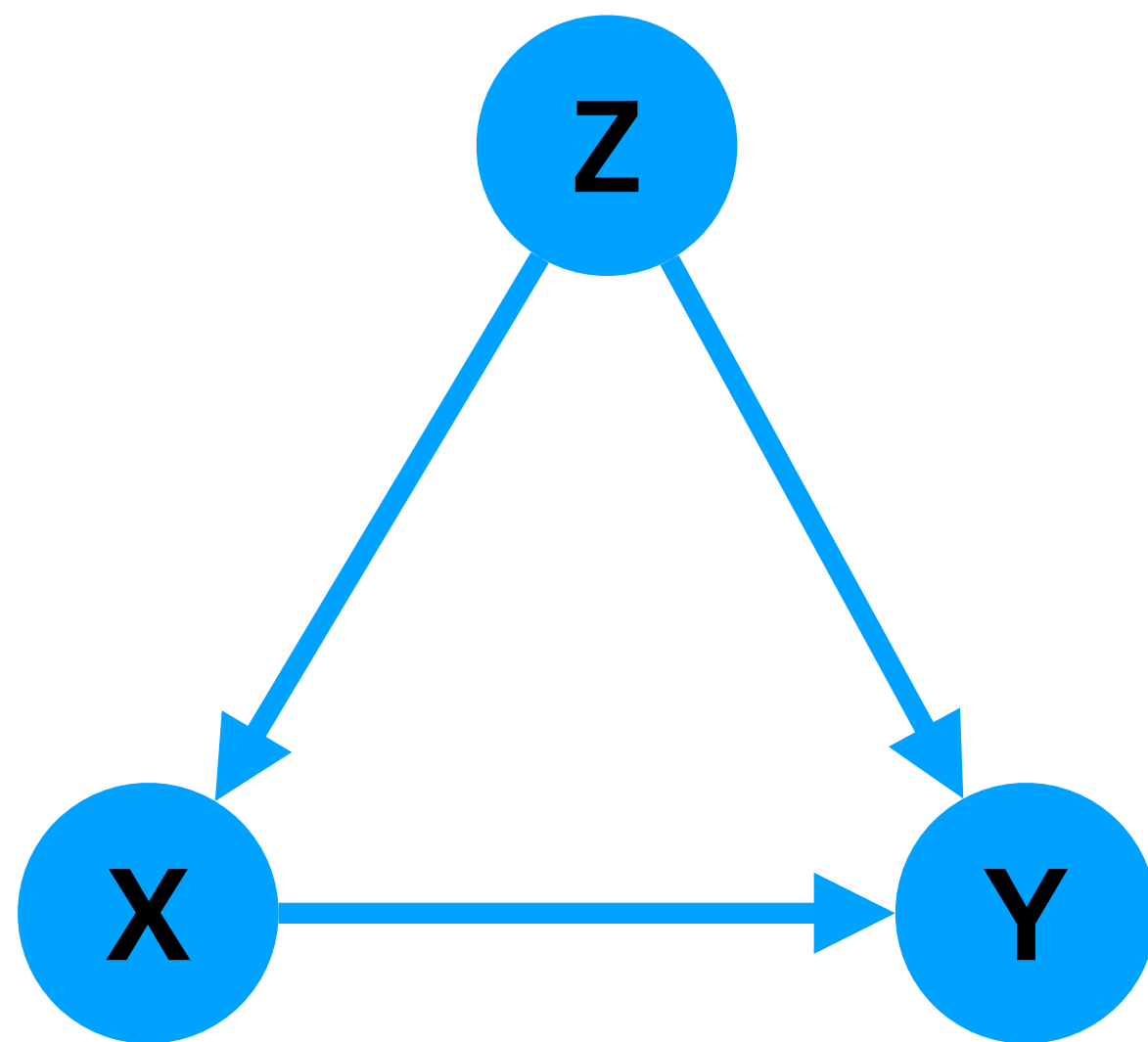
Paths and associations

**Arrows in a DAG
transmit associations**

**You can redirect and control those paths
by "adjusting" or "conditioning"**

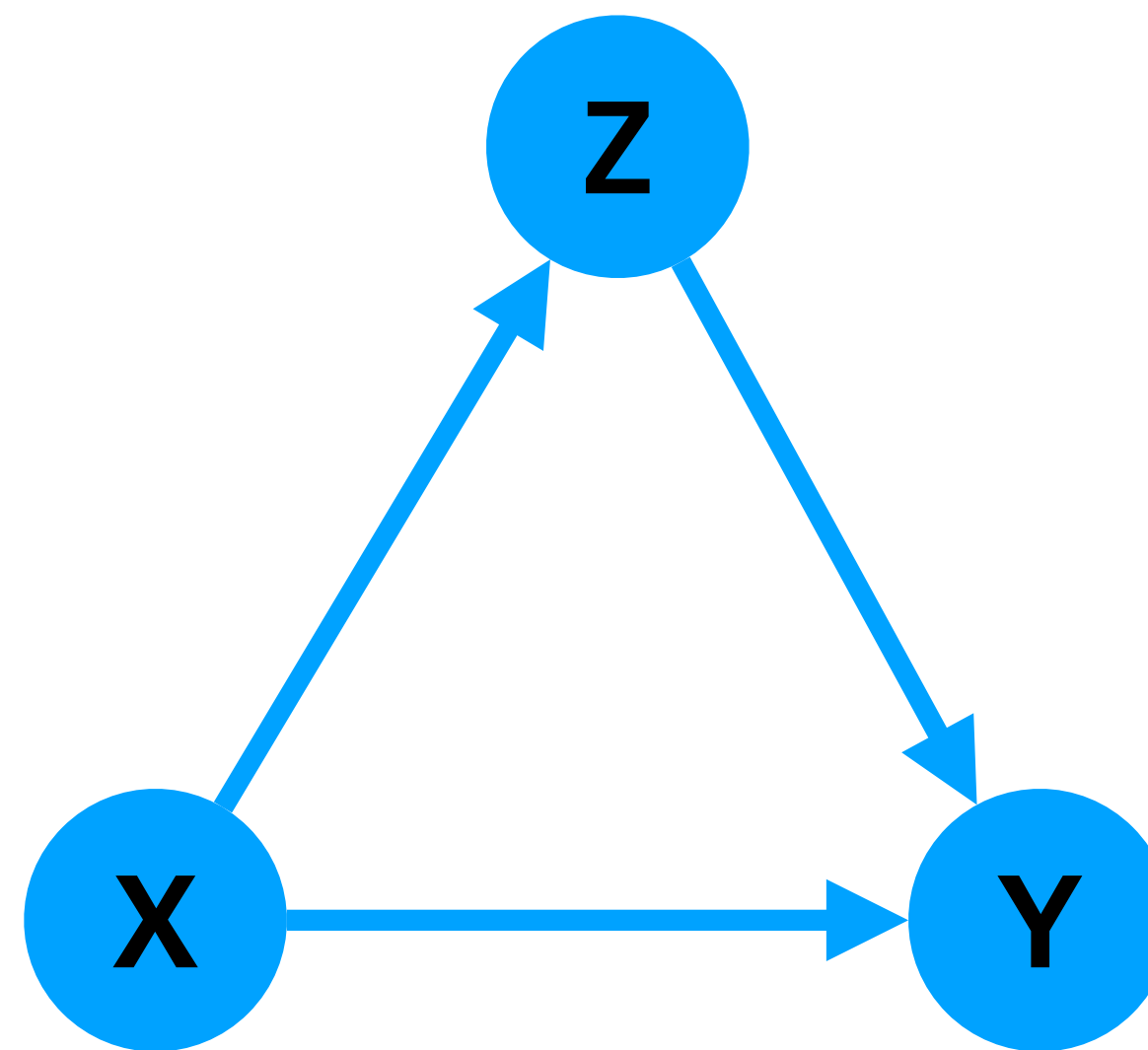
Three types of associations

Confounding



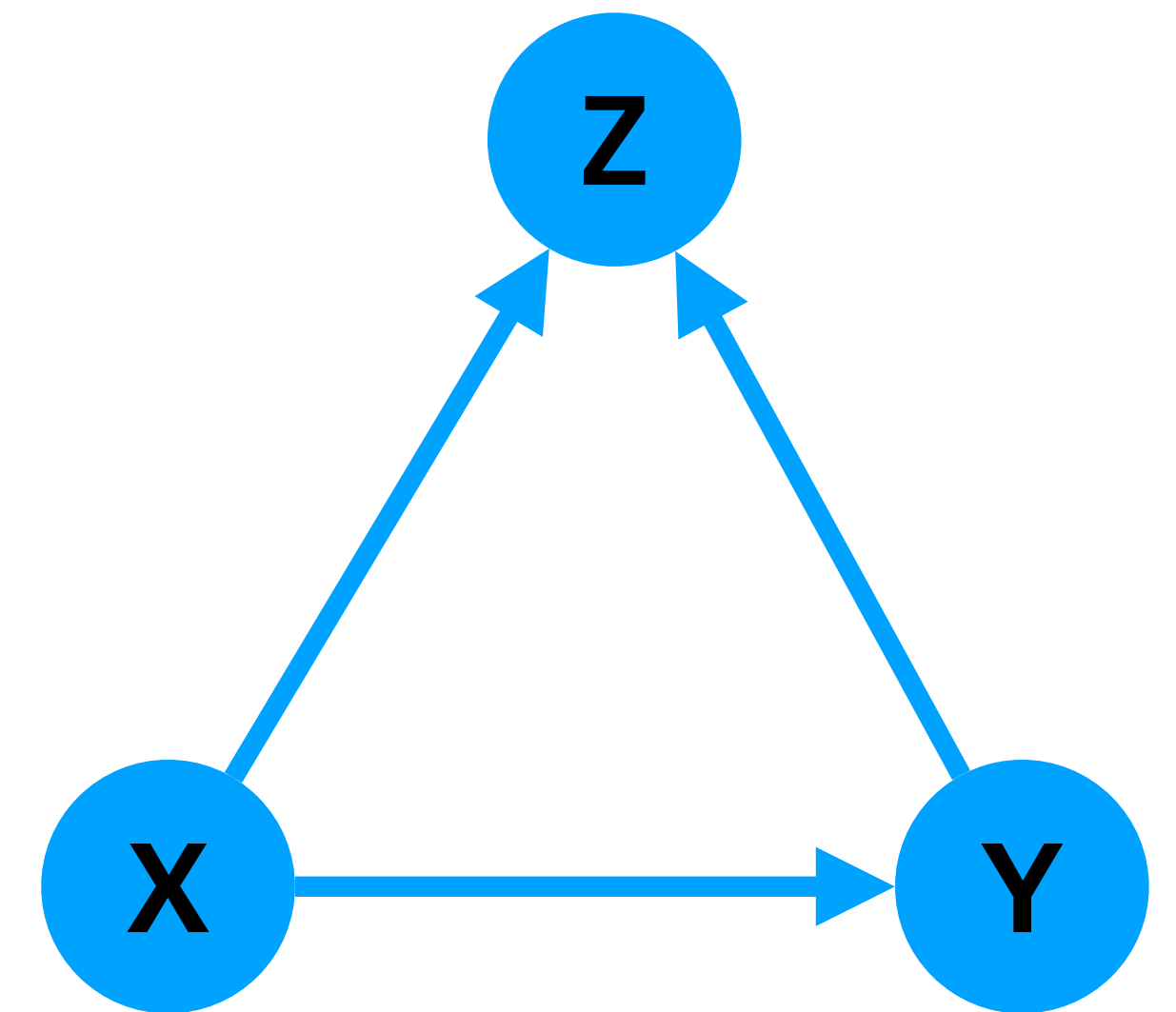
Common cause

Causation



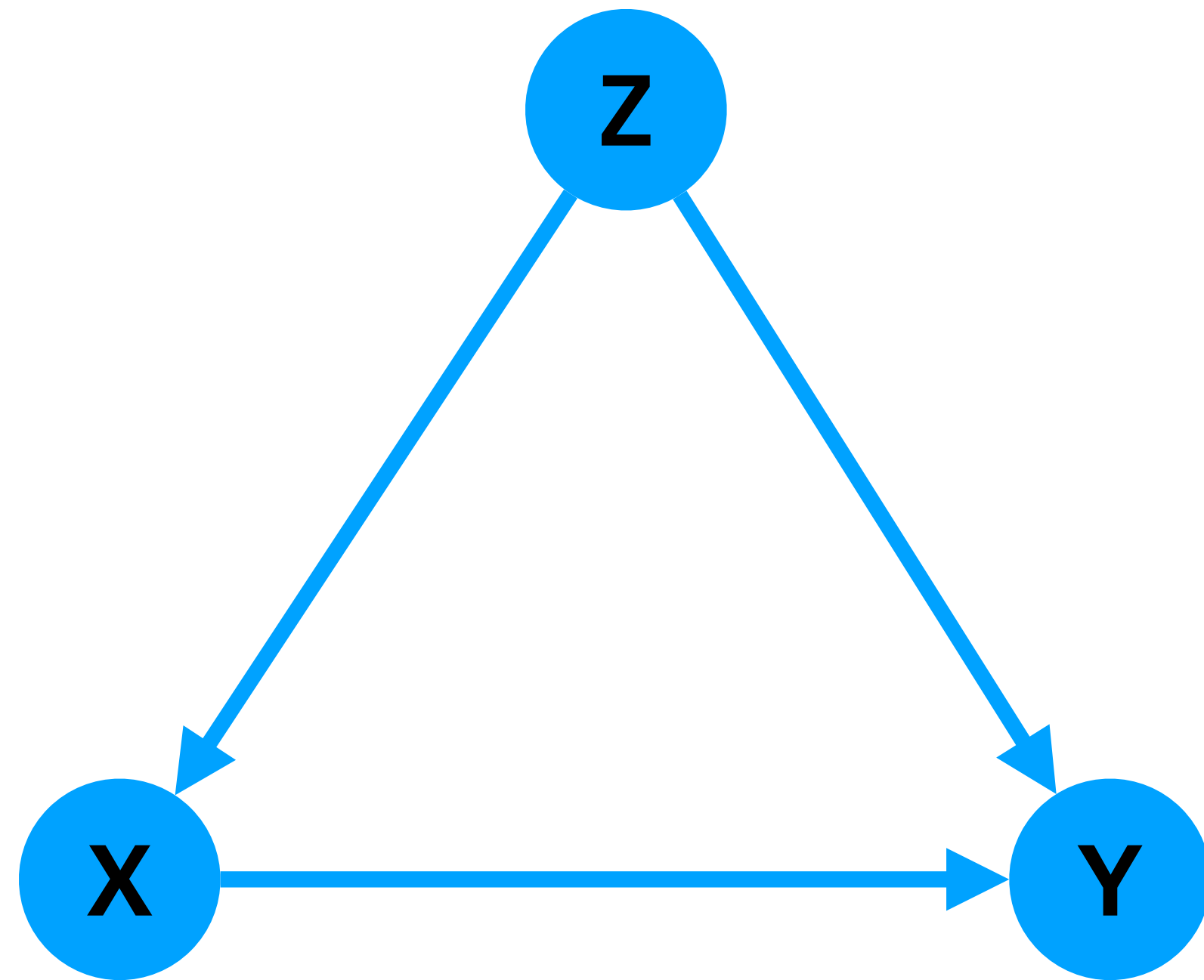
Mediation

Collision



Selection /
endogeneity

Confounding

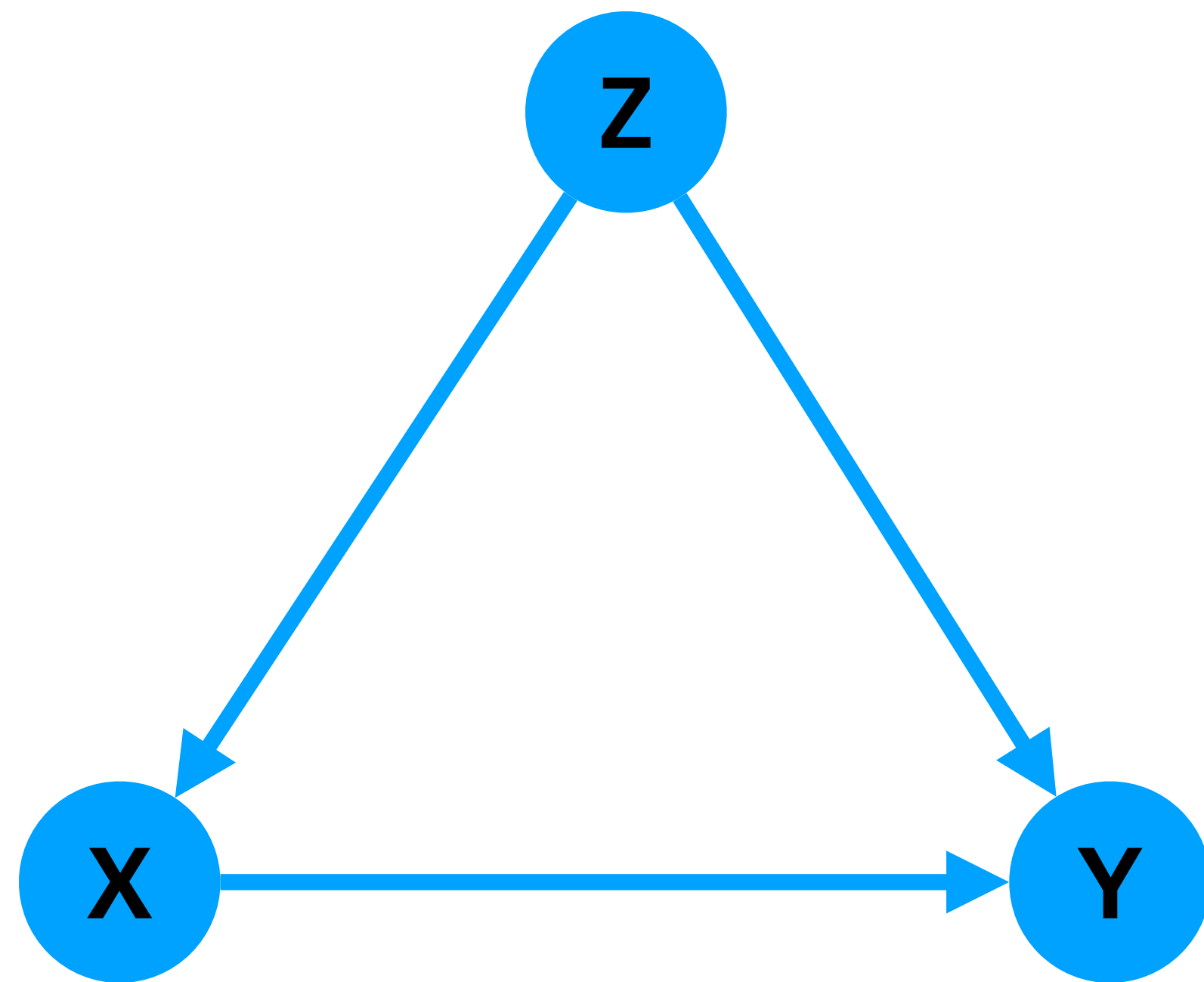


X causes Y

**But Z causes
both X and Y**

**Z confounds the
 $X \rightarrow Y$ association**

Paths



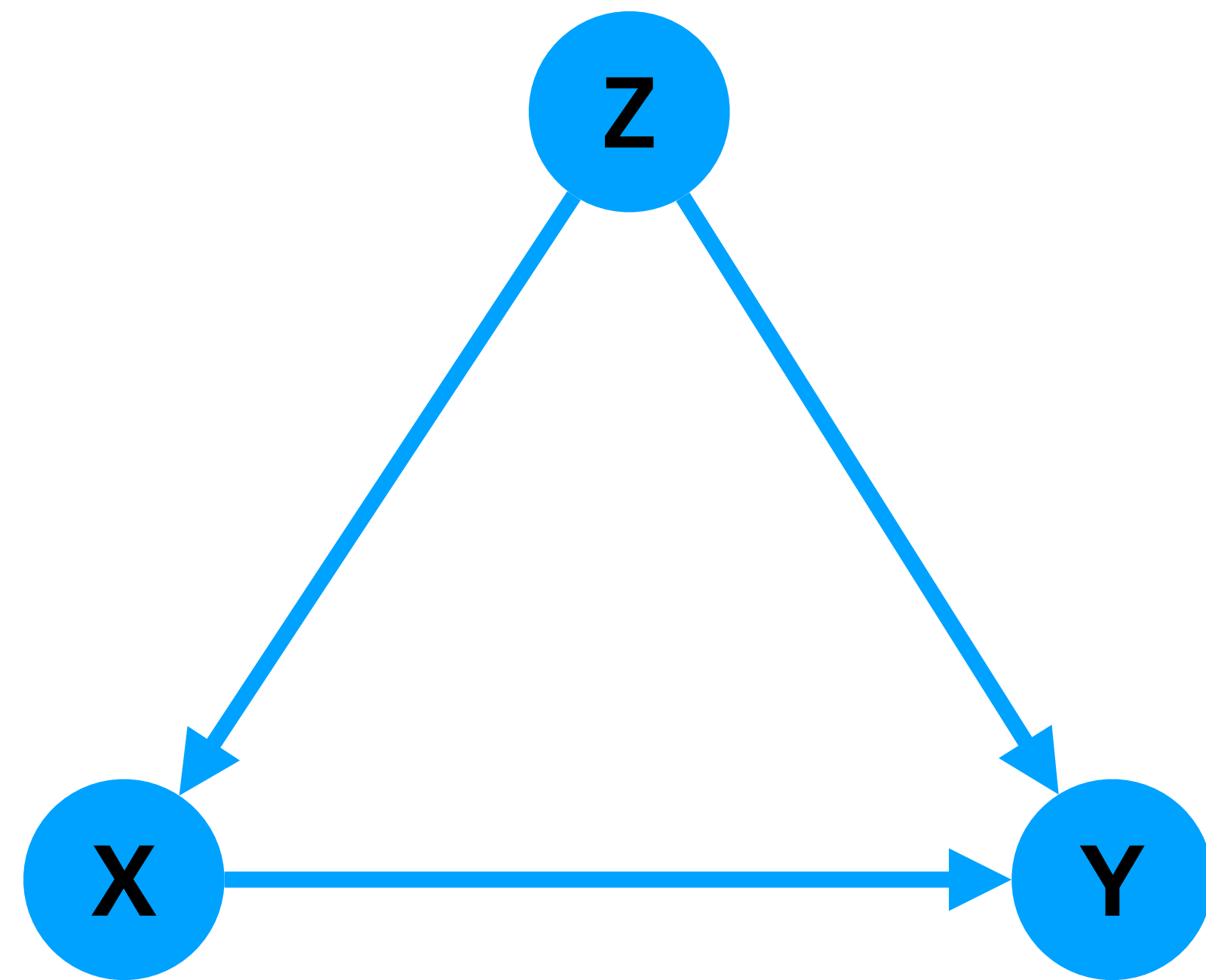
Paths between
X and Y?

$X \rightarrow Y$

$X \leftarrow Z \rightarrow Y$

Z is a
backdoor

d-connection

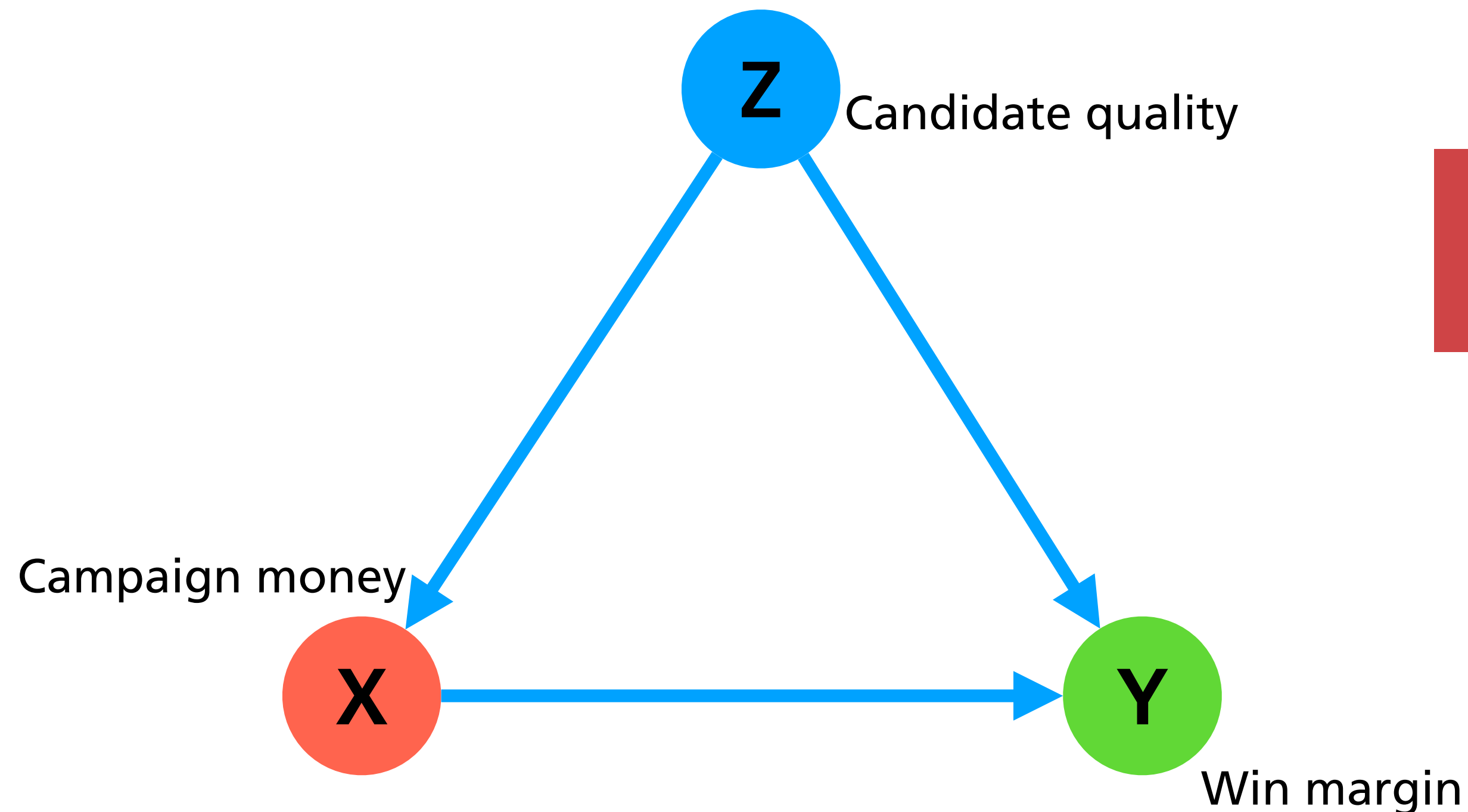


X and *Y* are "*d*-connected" because associations can pass through *Z*

The relationship between *X* and *Y* is not identified / isolated

Effect of money on elections

What are the paths between money and win margin?



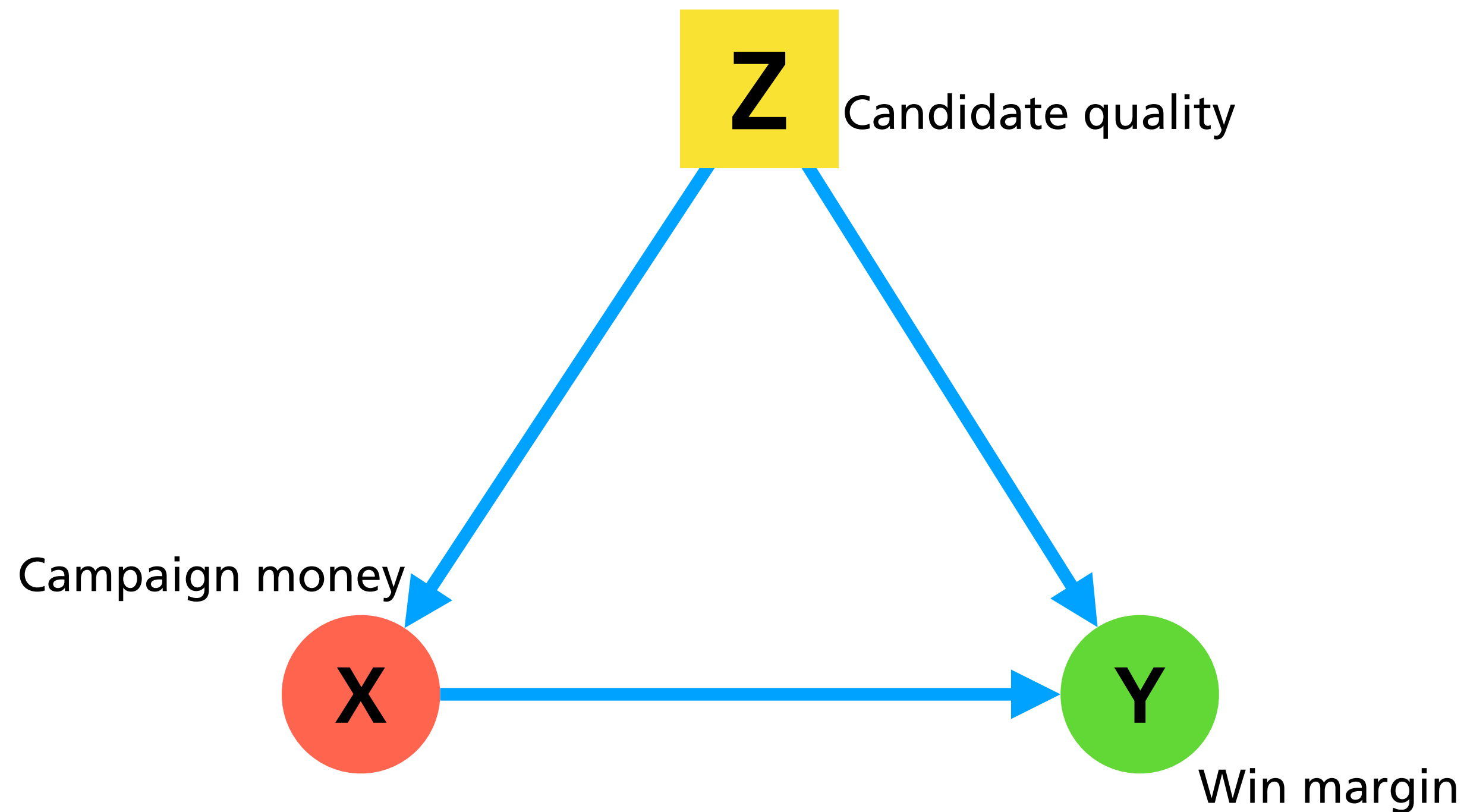
Money → Margin

Money ← Quality → Margin

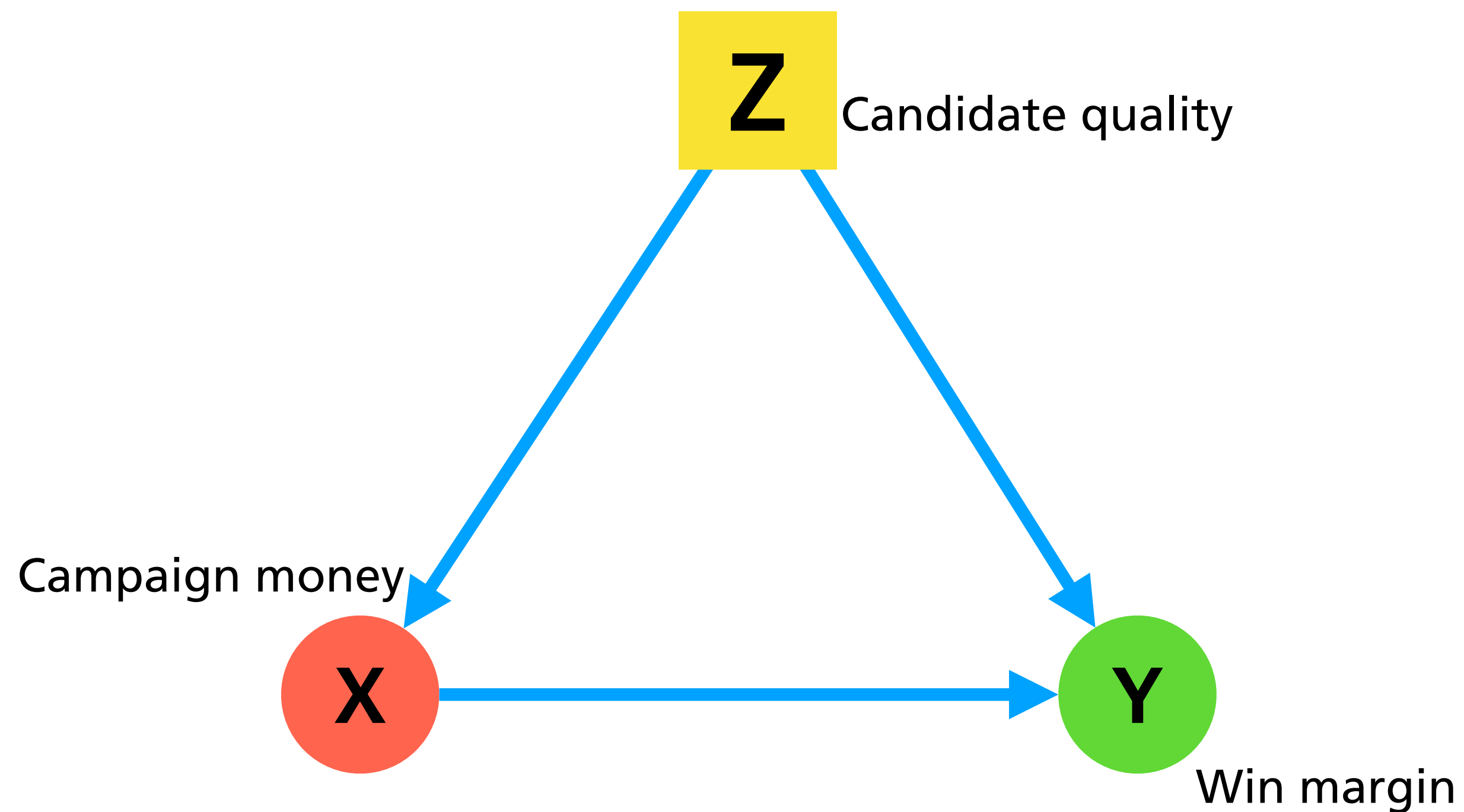
Quality is a *backdoor*

Closing doors

Close the backdoor
by adjusting for Z



Closing doors



**Close the backdoor
by adjusting for Z**

Find the part of campaign money that is explained by quality, remove it. This is the residual part of money.

Find the part of win margin that is explained by quality, remove it. This is the residual part of win margin.

Find the relationship between the residual part of money and residual part of win margin. **This is the causal effect.**

How to adjust

Include term in regression

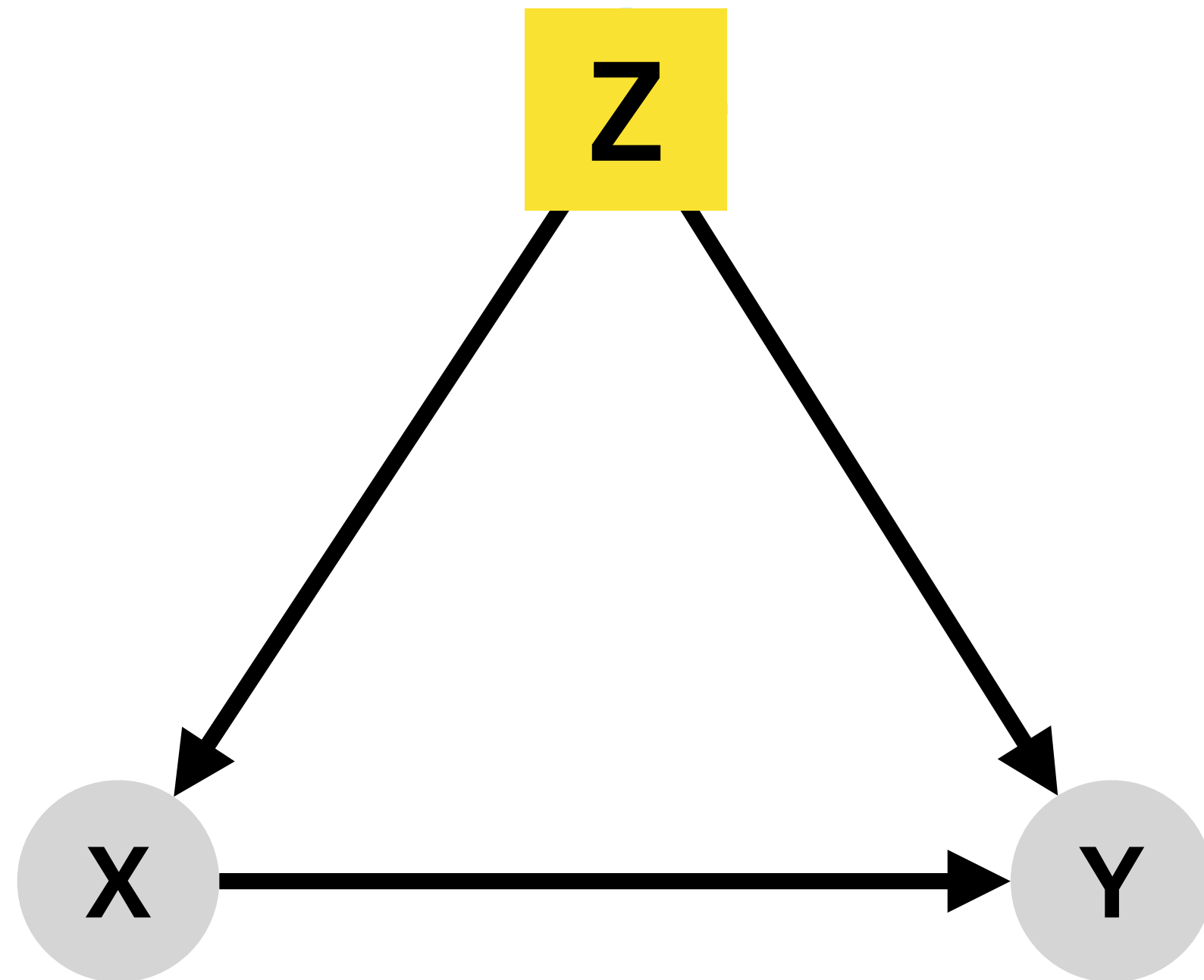
$$\text{Win margin} = \beta_0 + \beta_1 \text{Campaign money} + \beta_2 \text{Candidate quality} + \varepsilon$$

Matching

Stratifying

Inverse probability weighting

d-separation

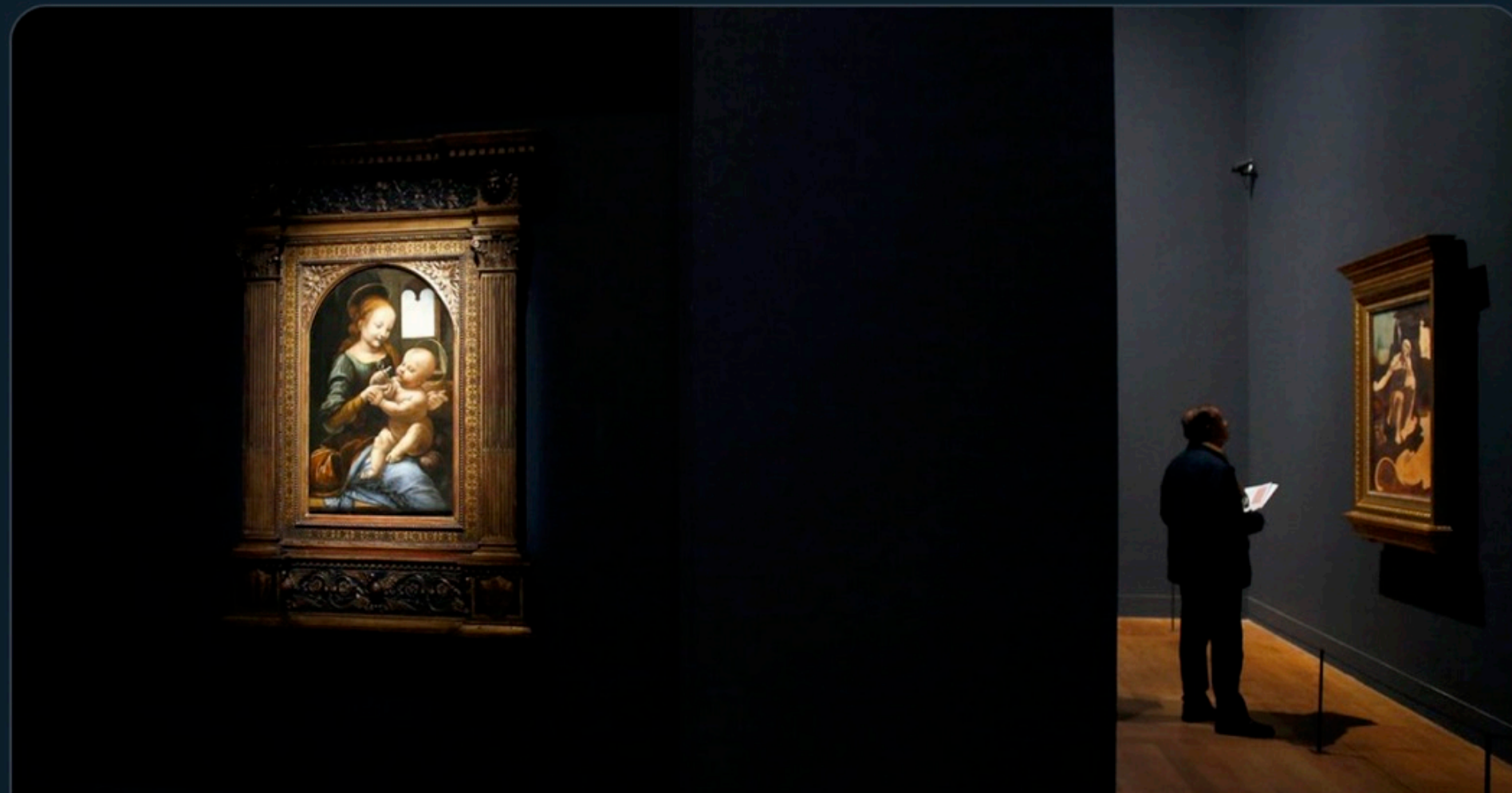


If we control for Z,
X and Y are now
"*d*-separated" and
the association is
isolated!



NYT Health
@NYTHealth

Want to live longer? Try going to the opera. Researchers in Britain have found that people who reported going to a museum or concert even once a year lived longer than those who didn't.



Another Benefit to Going to Museums? You May Live Longer

Researchers in Britain found that people who go to museums, the theater and the opera were less likely to die in the study period than those who didn't.

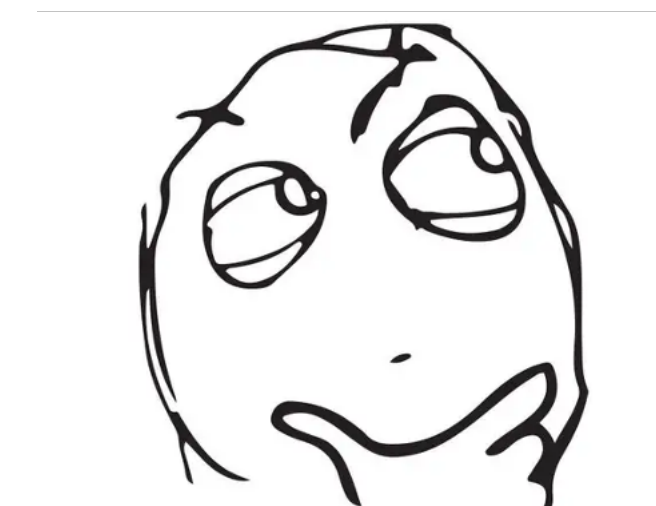
[nytimes.com](https://www.nytimes.com)

9:19 AM · Dec 22, 2019 · [SocialFlow](#)

336 Retweets 1.3K Likes

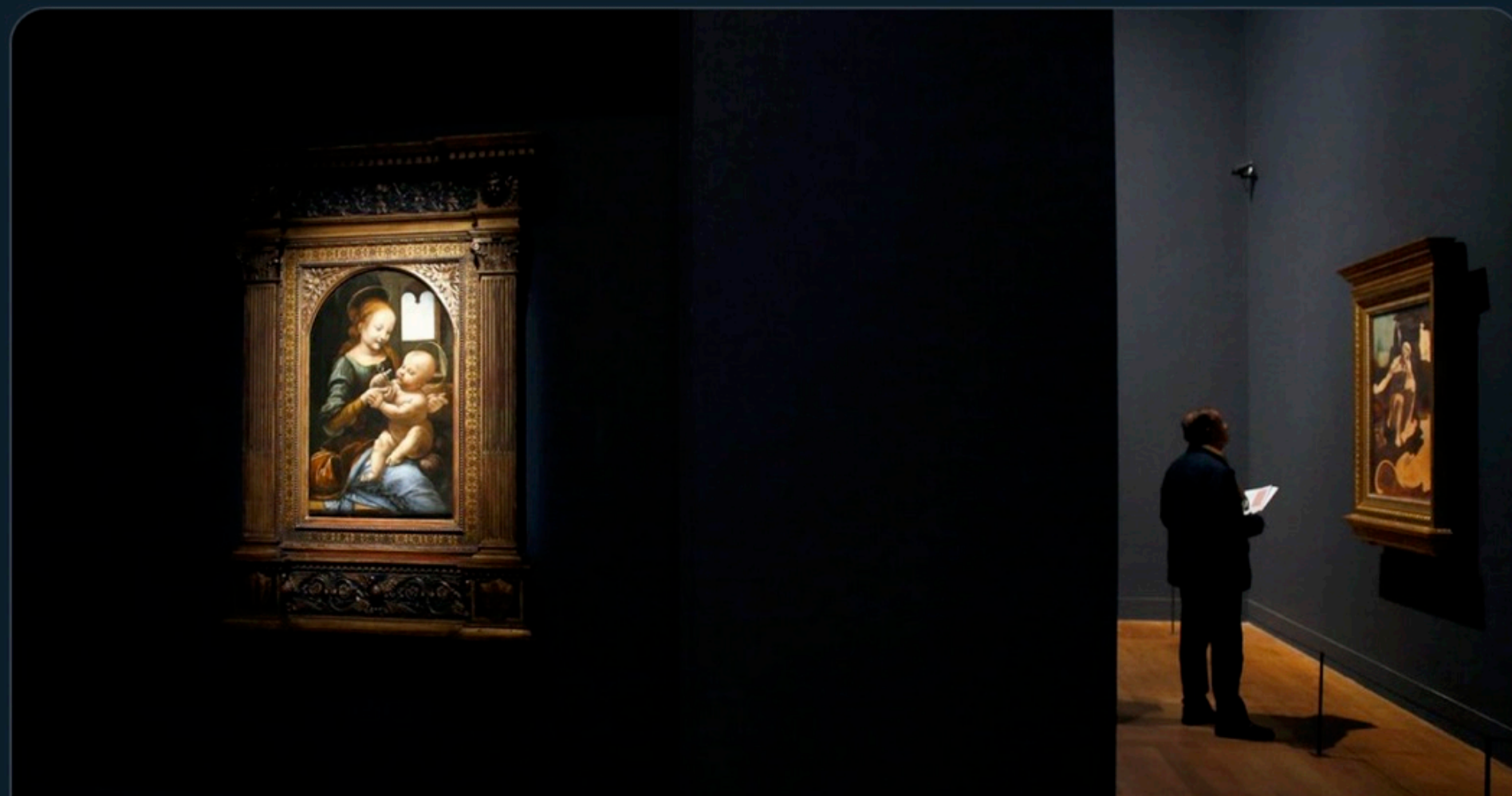


Something
wrong?





Want to live longer? Try going to the opera. Researchers in Britain have found that people who reported going to a museum or concert even once a year lived longer than those who didn't.



Another Benefit to Going to Museums? You May Live Longer

Researchers in Britain found that people who go to museums, the theater and the opera were less likely to die in the study period than those who didn't.

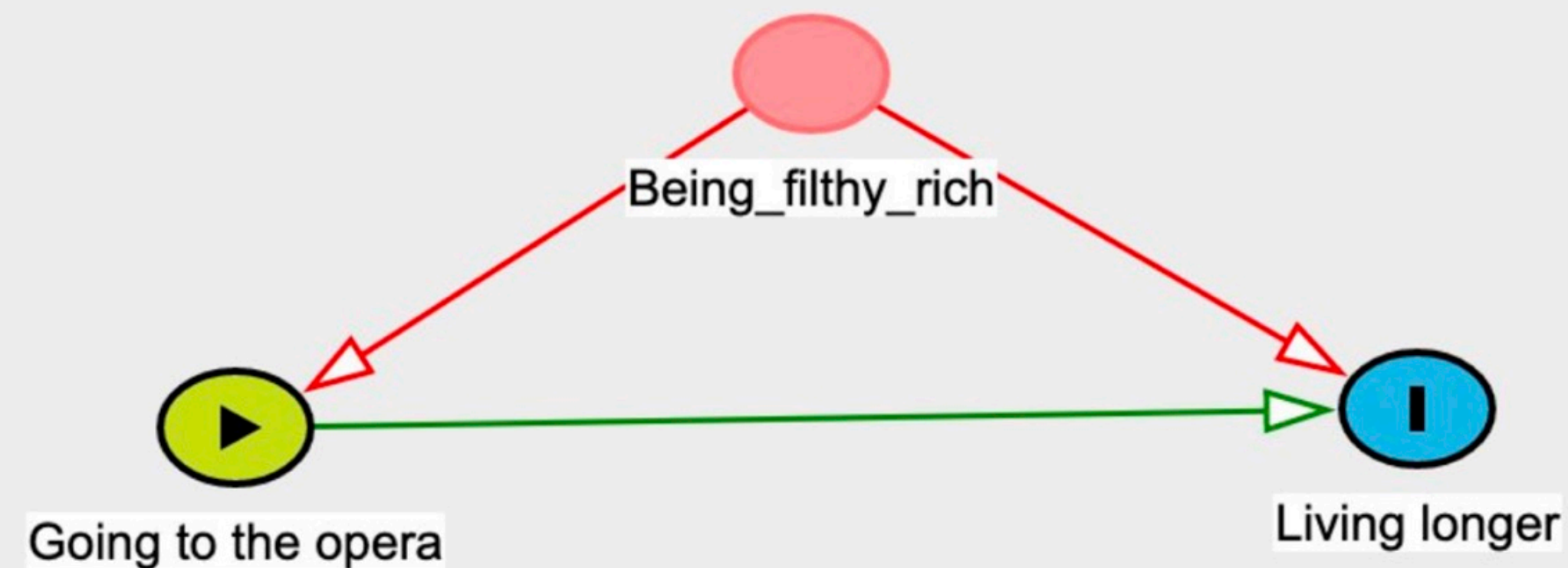
[nytimes.com](https://www.nytimes.com)

9:19 AM · Dec 22, 2019 · [SocialFlow](#)

336 Retweets 1.3K Likes



ooh ooh i can draw the dag for this one!



NYT Health @NYTHealth · Dec 22, 2019

Want to live longer? Try going to the opera. Researchers in Britain have found that people who reported going to a museum or concert even once a year lived longer than those who didn't. nyti.ms/2Q9AmZV

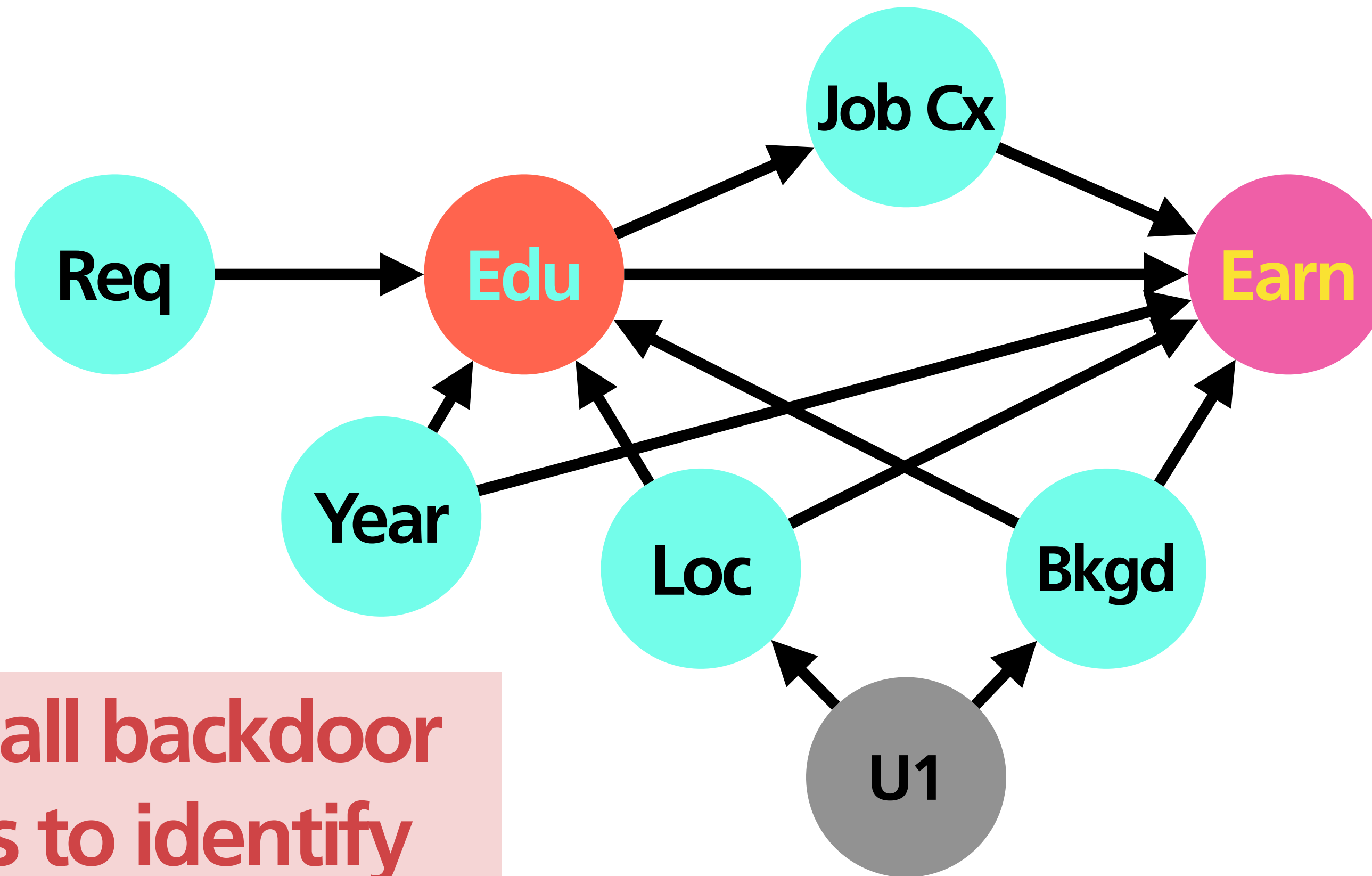
2:47 PM · Dec 22, 2019 · [Twitter Web App](#)

View Tweet activity

837 Retweets 3.9K Likes

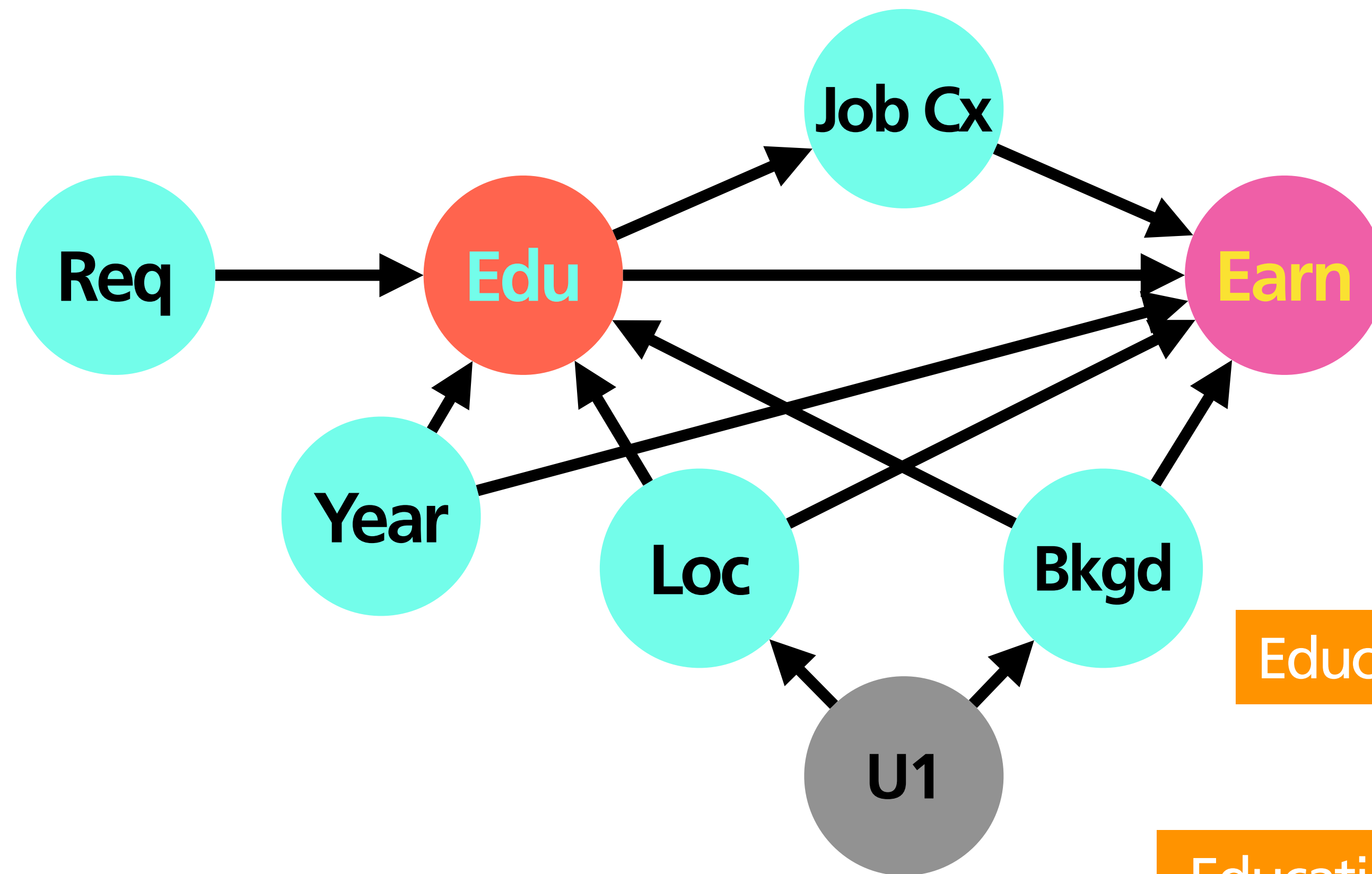
48 / 63

Closing backdoors



Block all backdoor paths to identify the main pathway you care about

All paths



Education → Earnings

Education → Job connections → Earnings

Education ← Background → Earnings

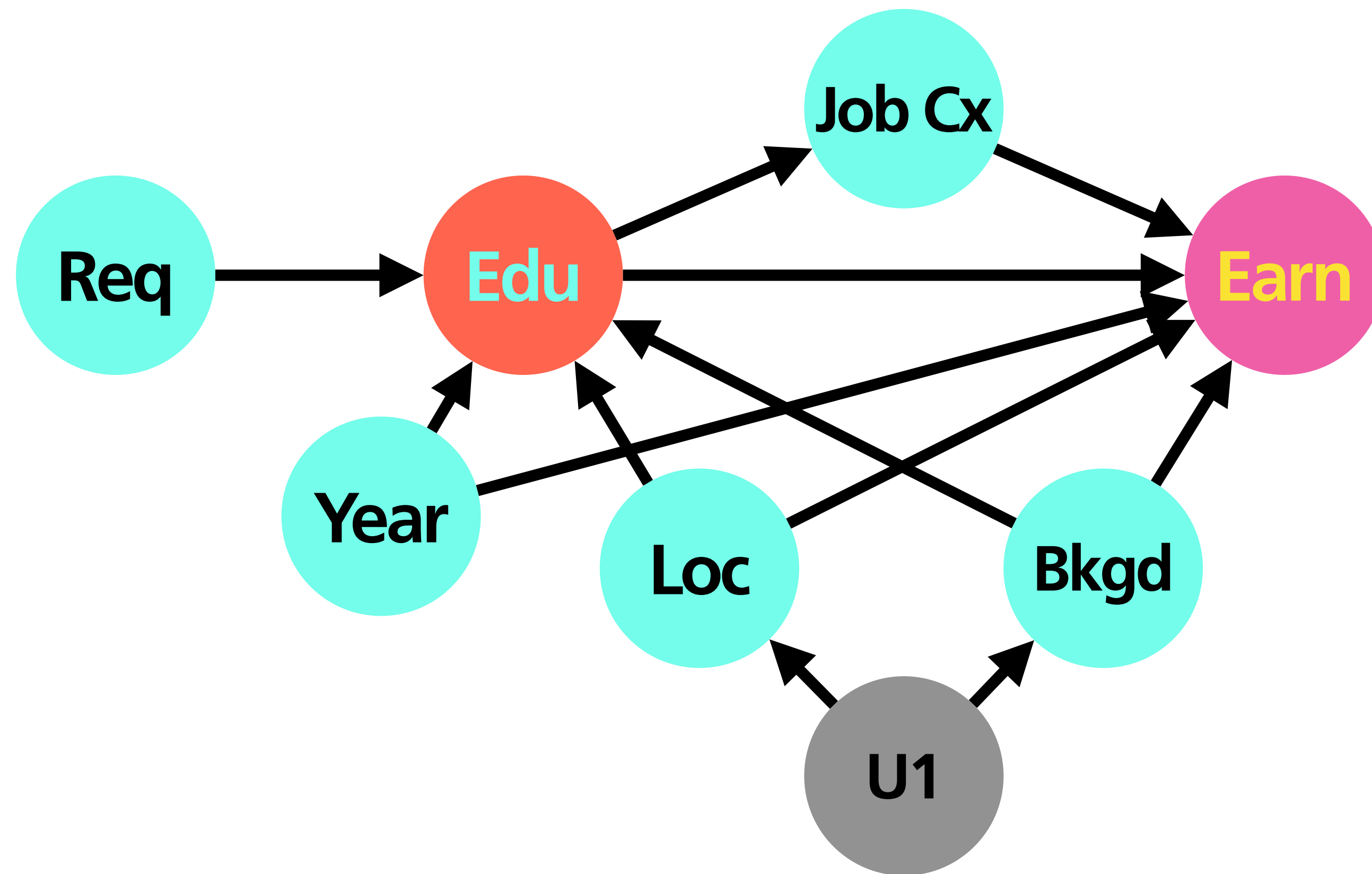
Education ← Location → Earnings

Education ← Background ← U1 → Location → Earnings

Education ← Year → Earnings

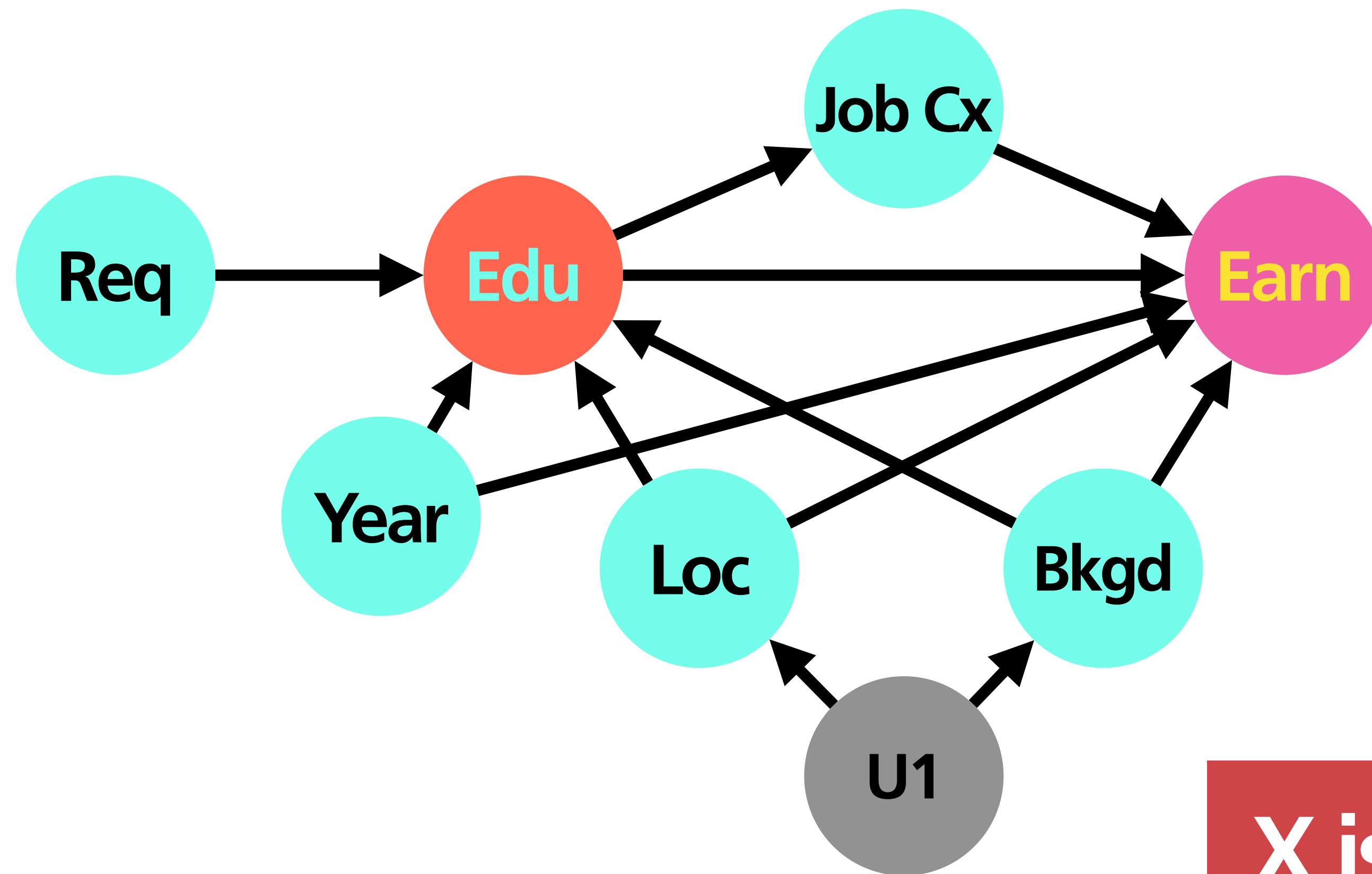
Education ← Location ← U1 → Background → Earnings

All paths



Adjust for Location,
Background and Year to isolate
the **Education** → **Earnings**
causal effect

How do you know if this is right?

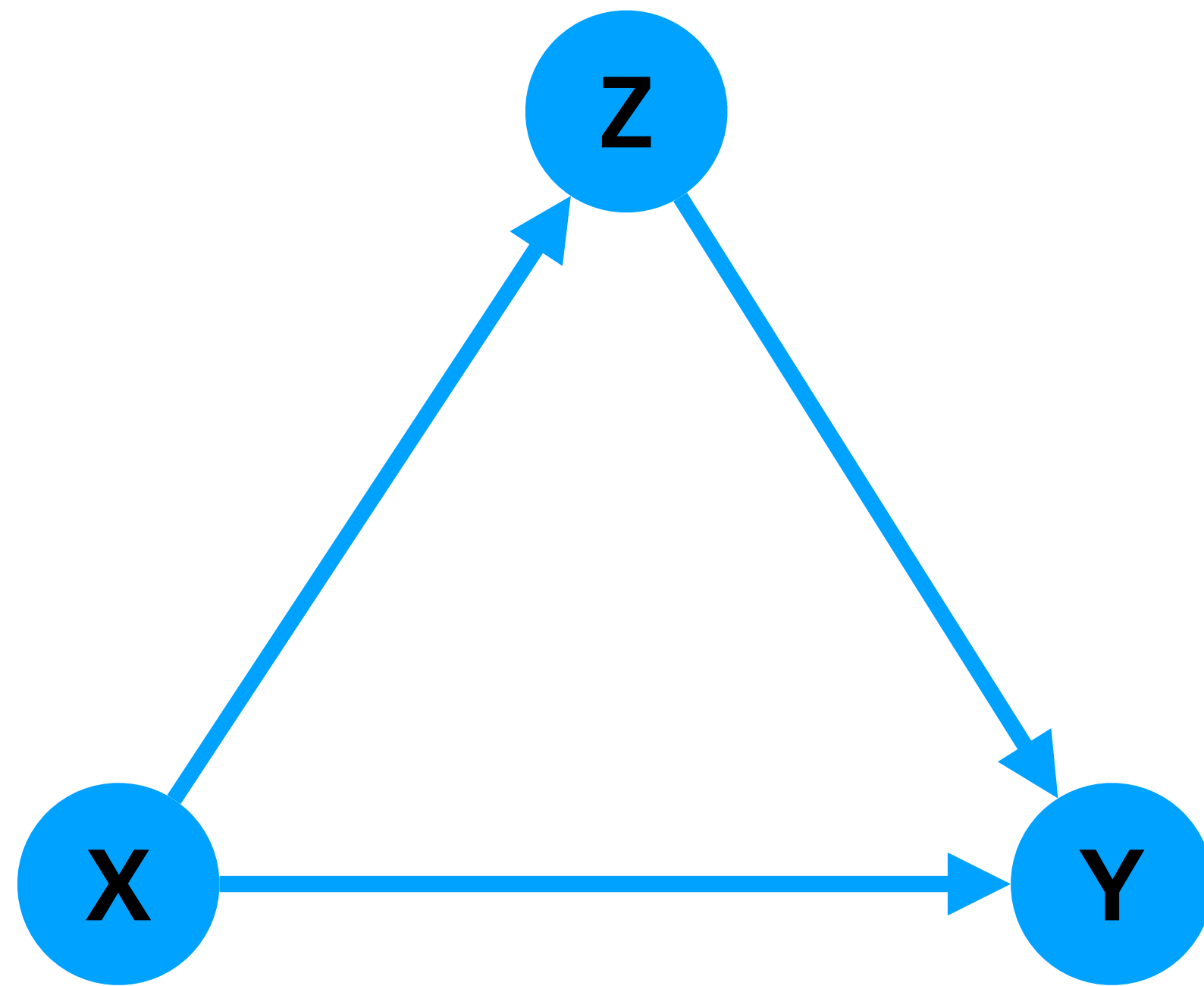


You can test the implications of the model to see if they're right in your data

$$X \perp Y | Z$$

X is independent of Y, given Z

Causation

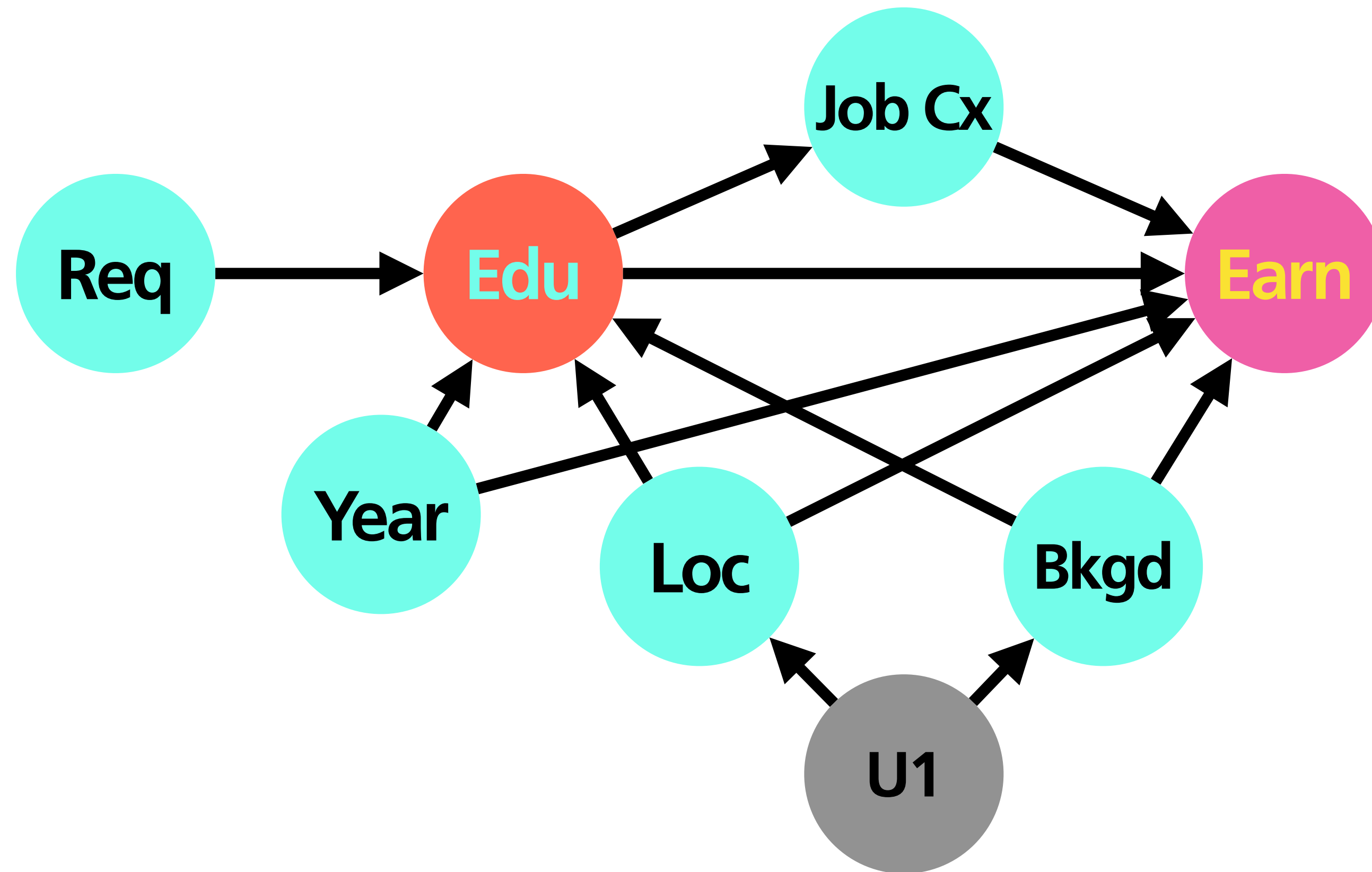


Should you
control for Z?

No!

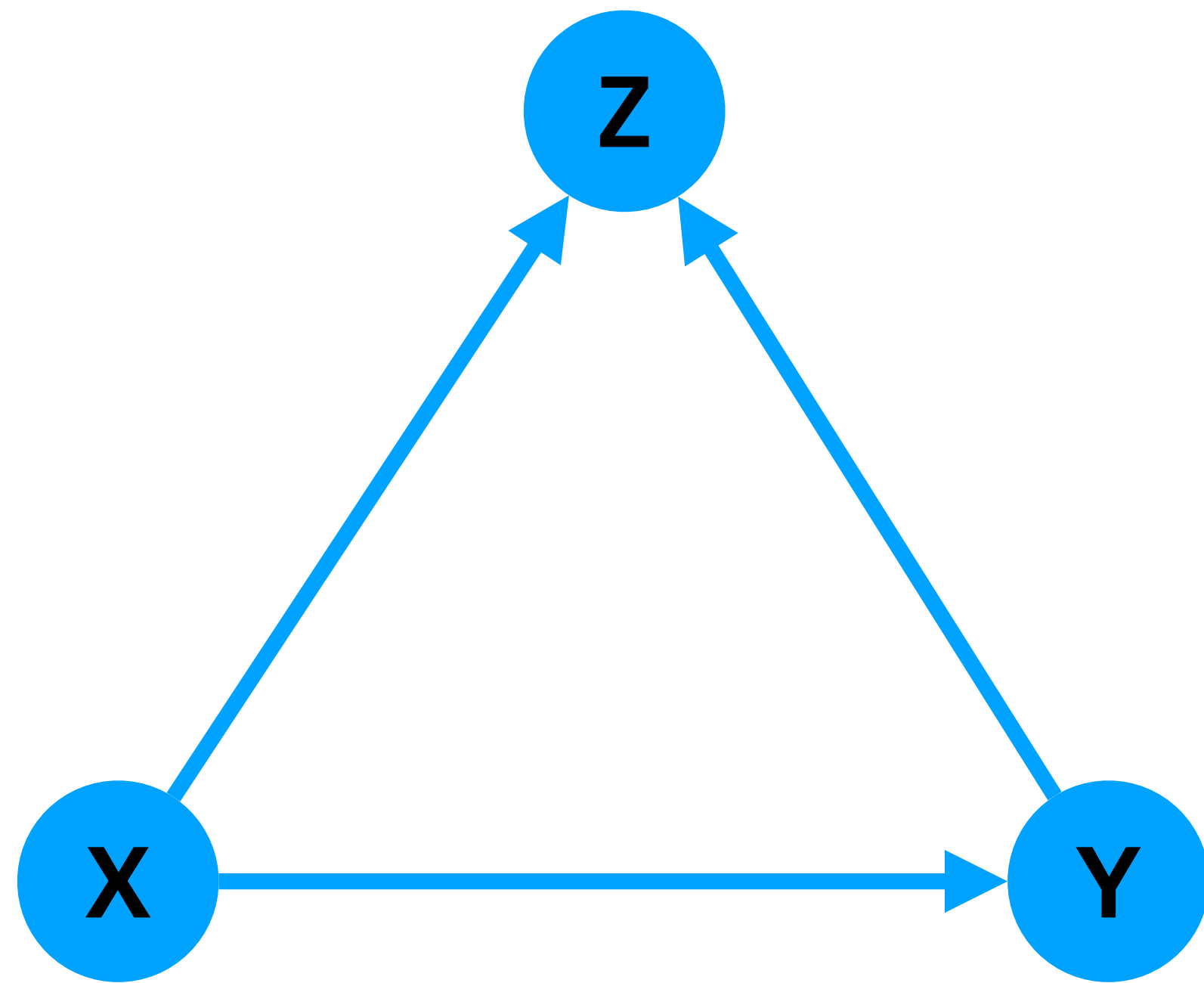
Overcontrolling

Causation and overcontrolling



Should you control for job connections?

Colliders



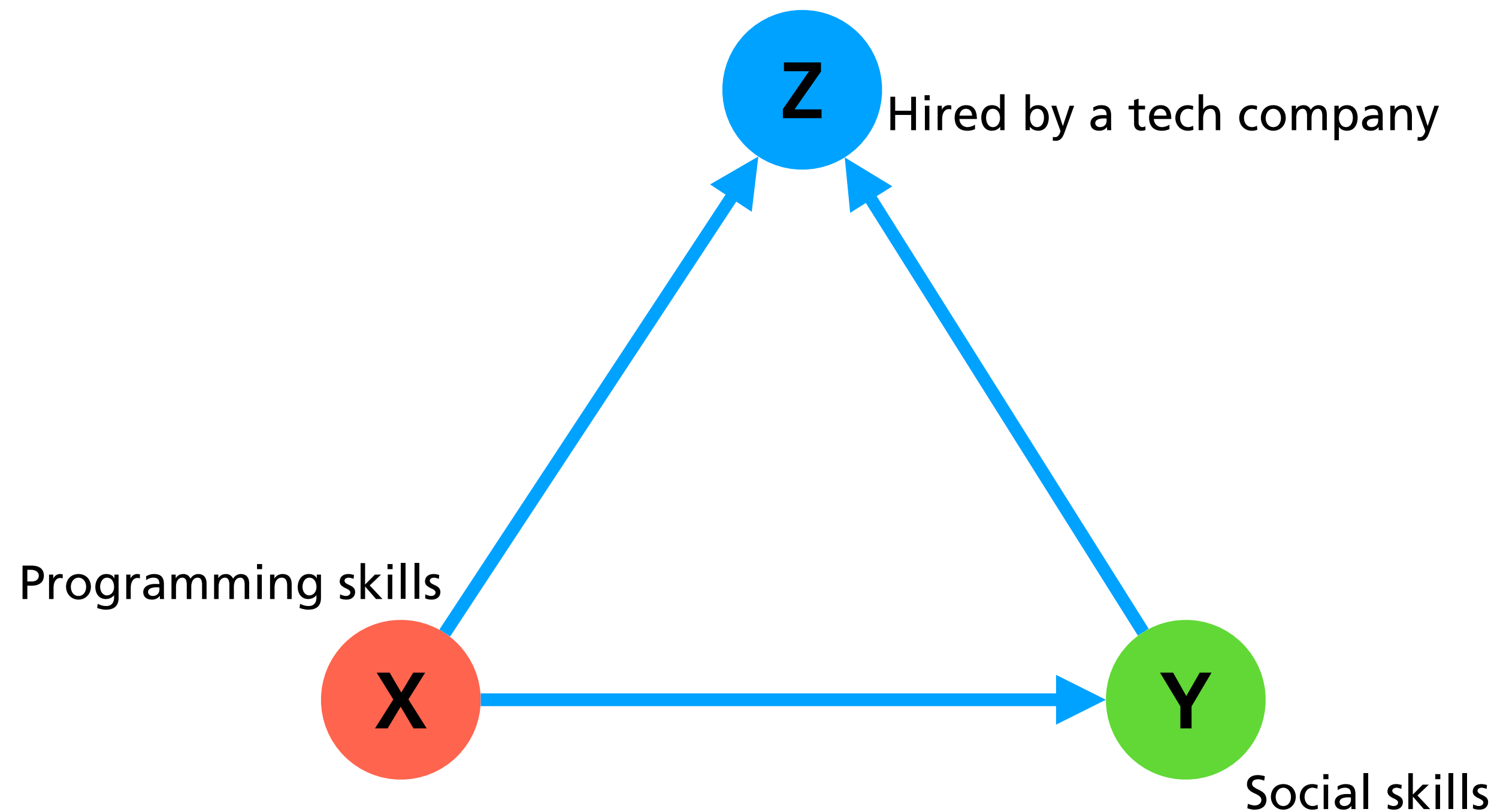
X causes Z

Y causes Z

Should you
control for Z?

Programming and social skills

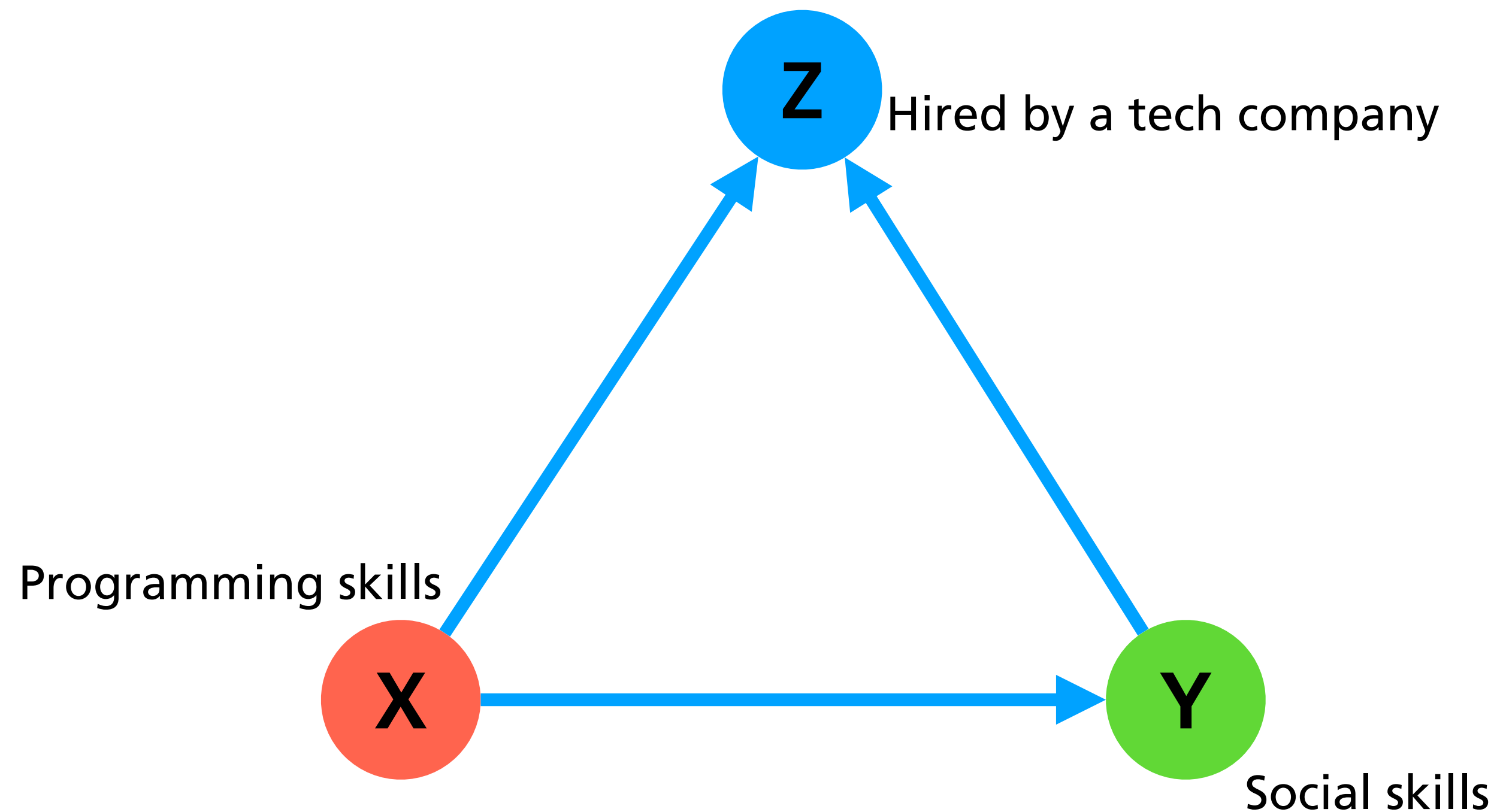
Do programming skills reduce social skills?



You go to a tech company and conduct a survey. You find a negative relationship! Is it real?

Programming and social skills

Do programming skills reduce social skills?

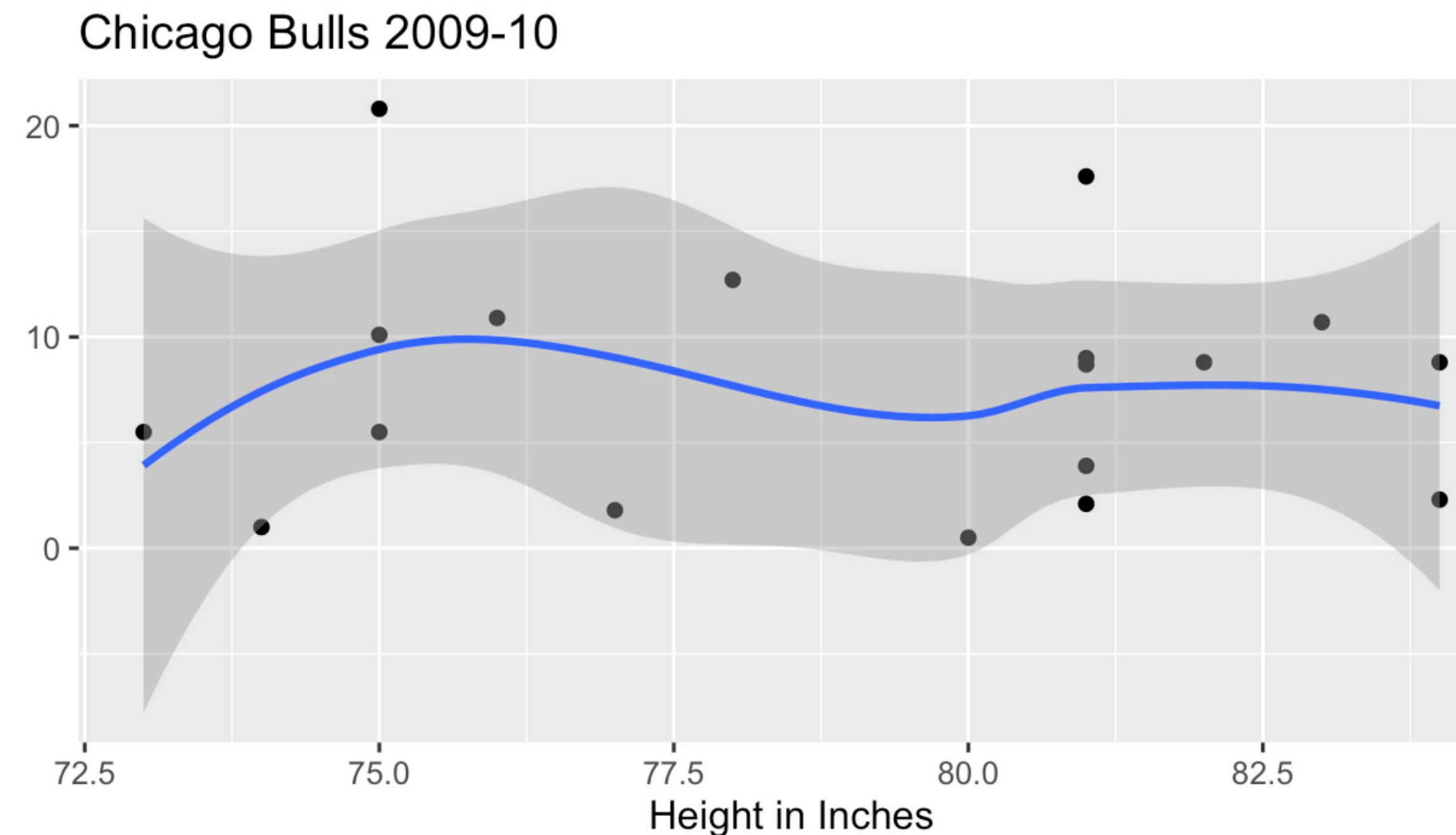


No! Hired by a tech company is a collider and we controlled for it.

This inadvertently connected the two.

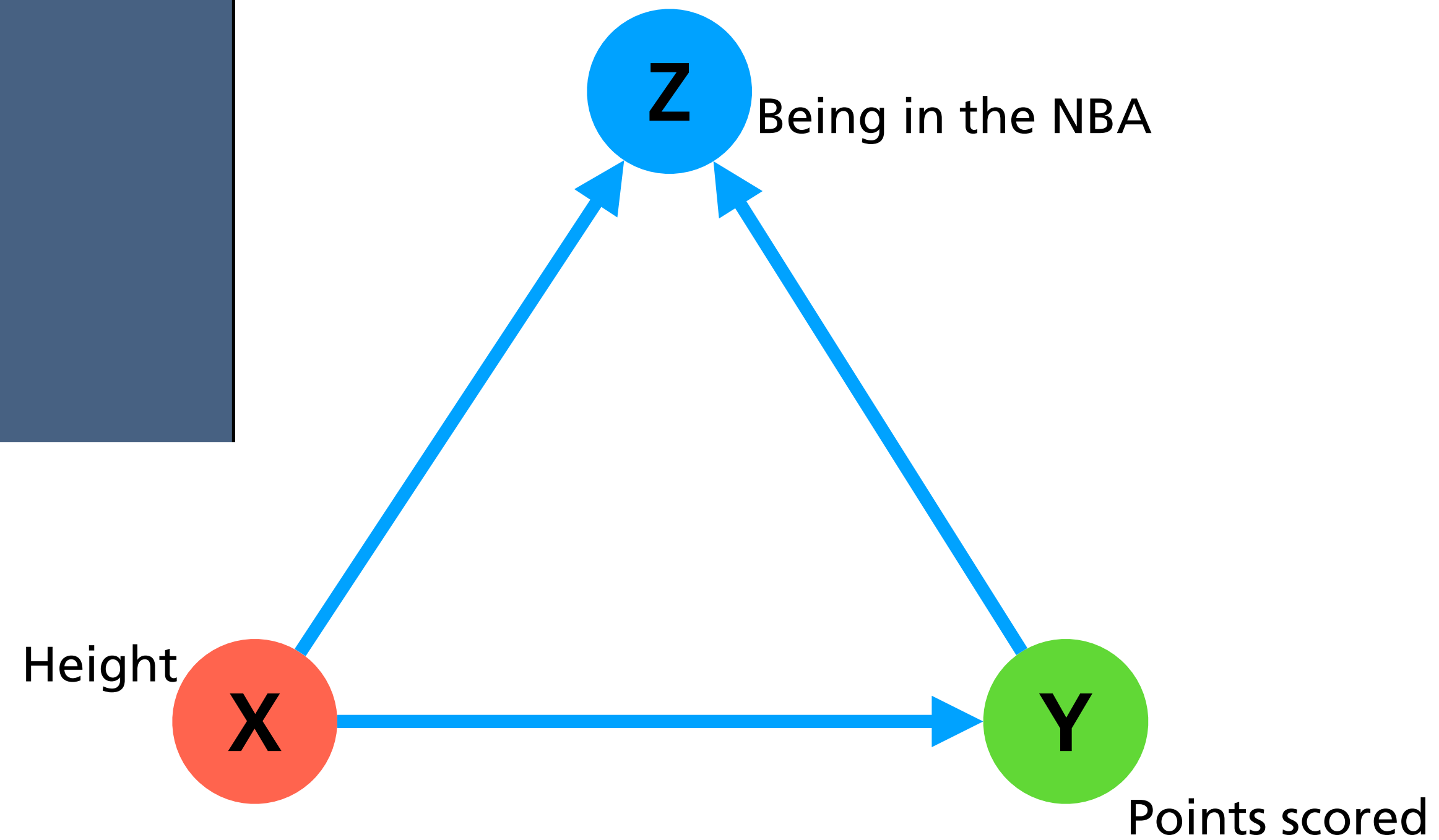
**Colliders can create
fake causal effects**

**Colliders can hide real
causal effects**

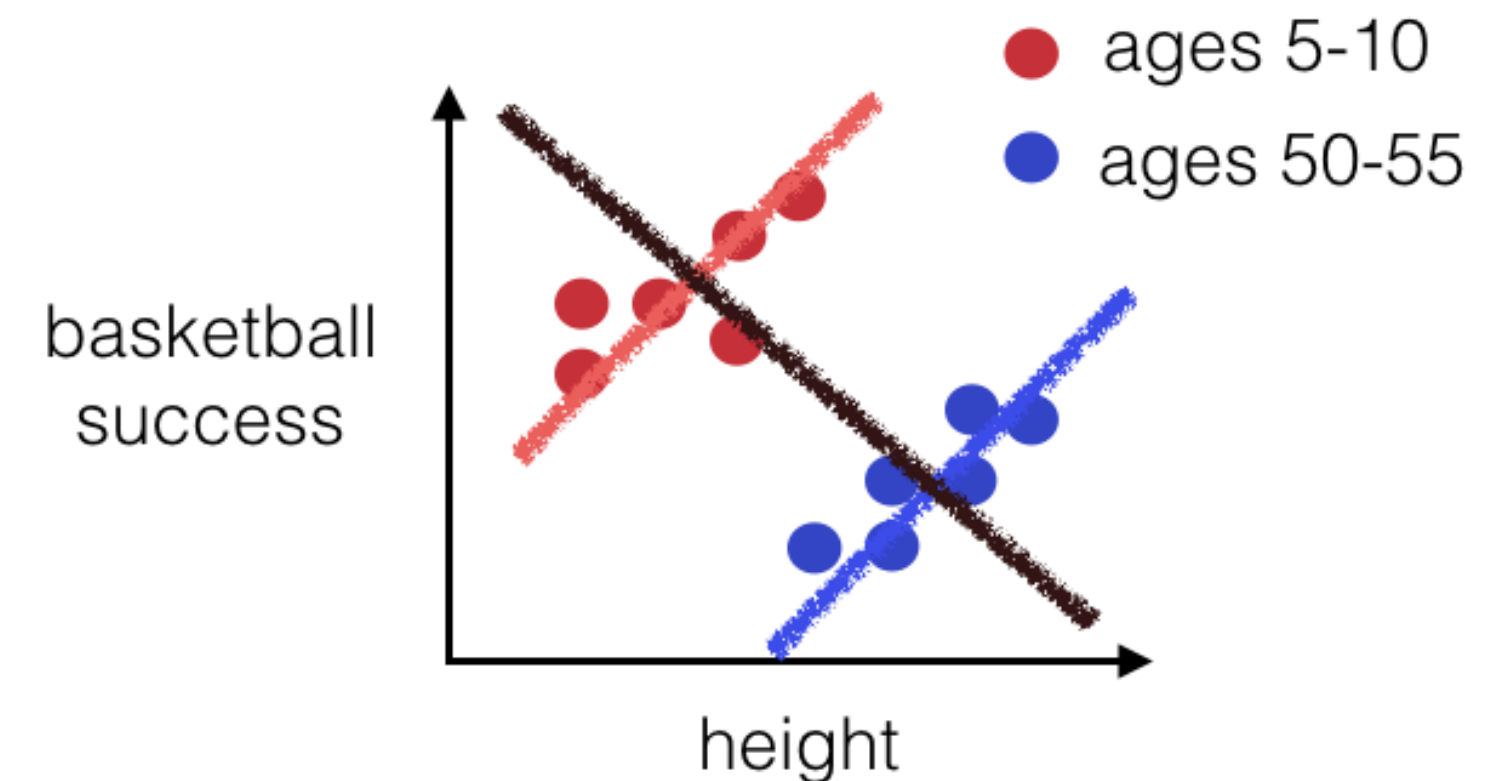
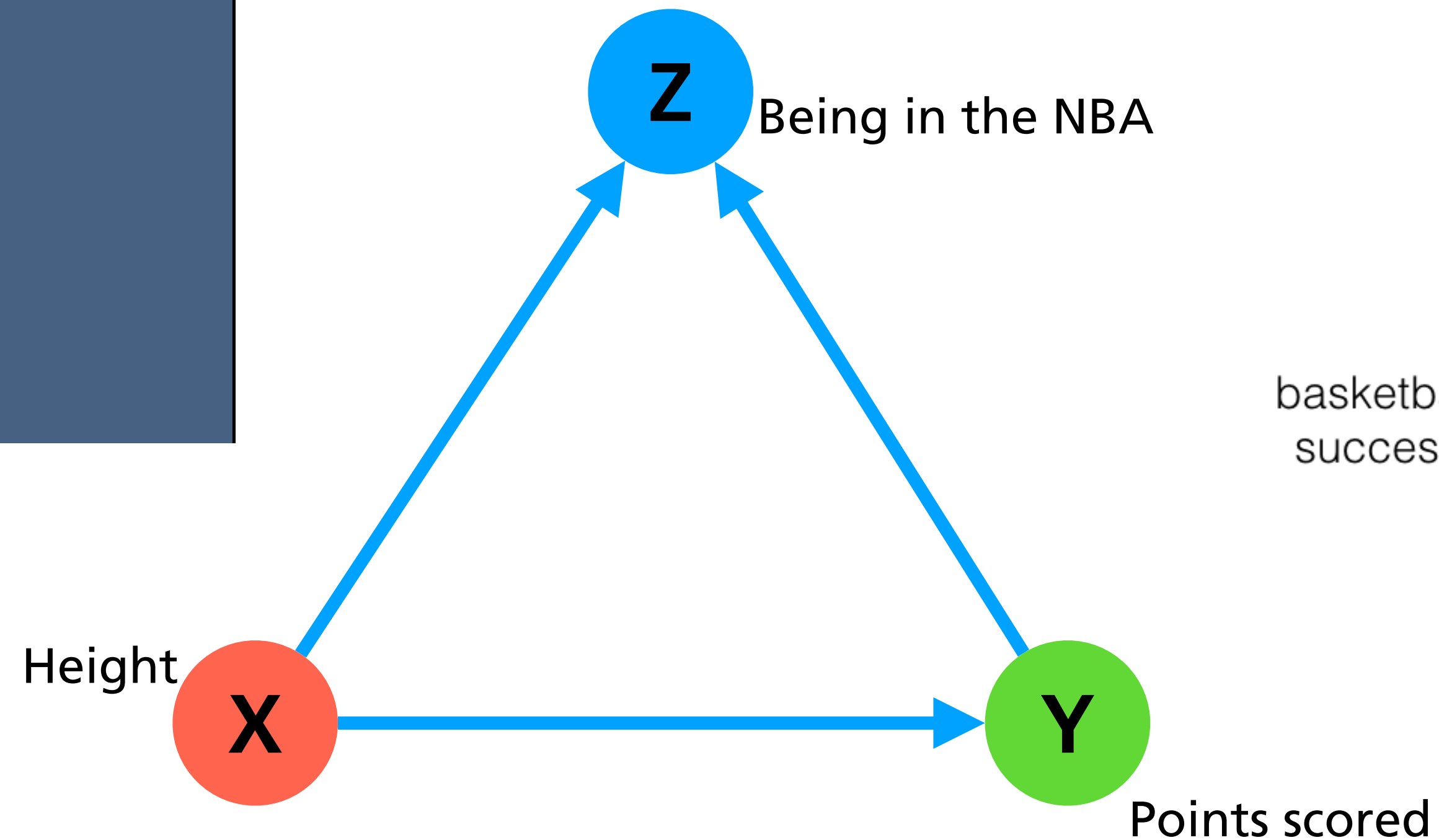


Height is unrelated to basketball skill... among NBA players

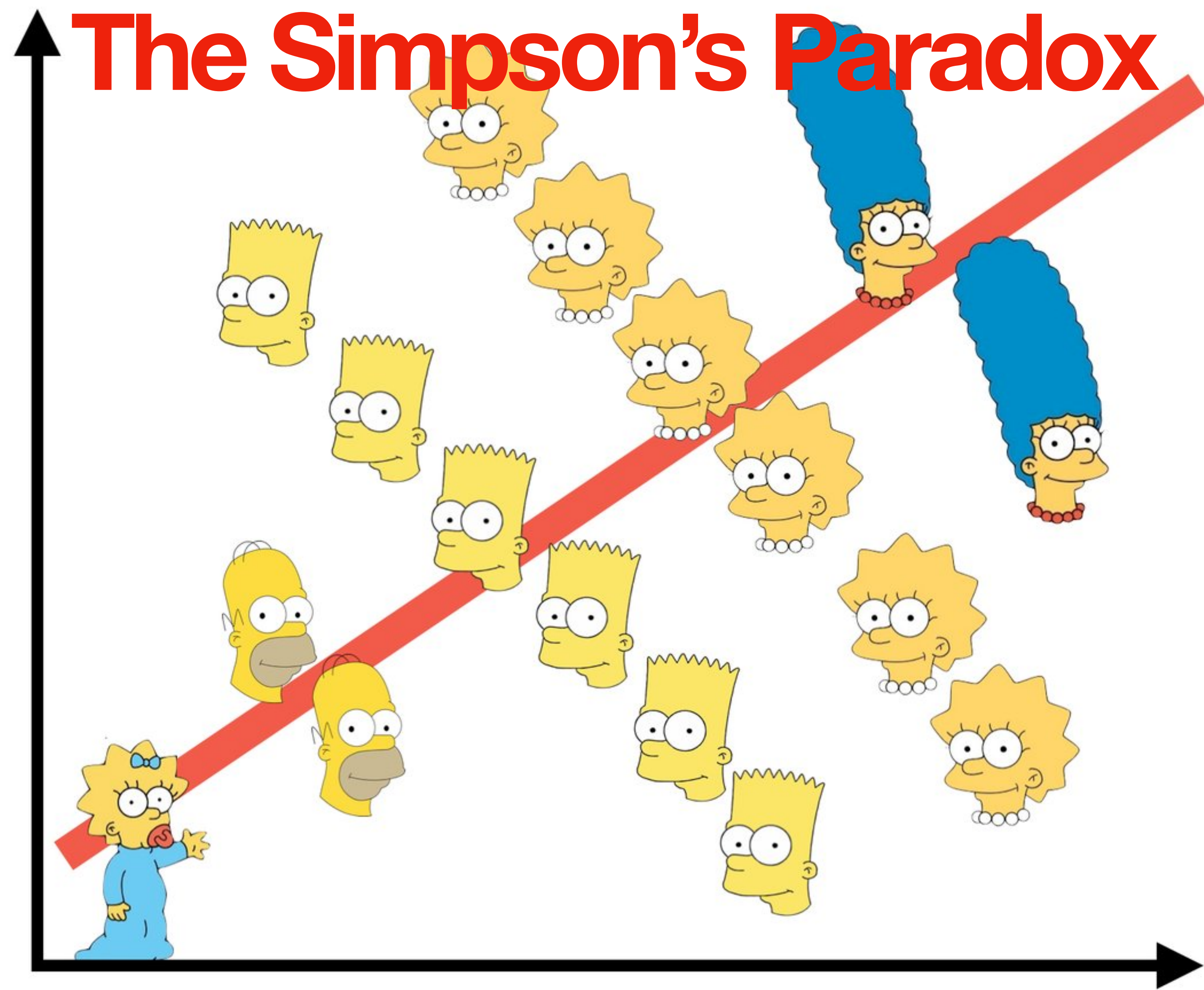
Colliders and selection bias



Colliders and selection bias

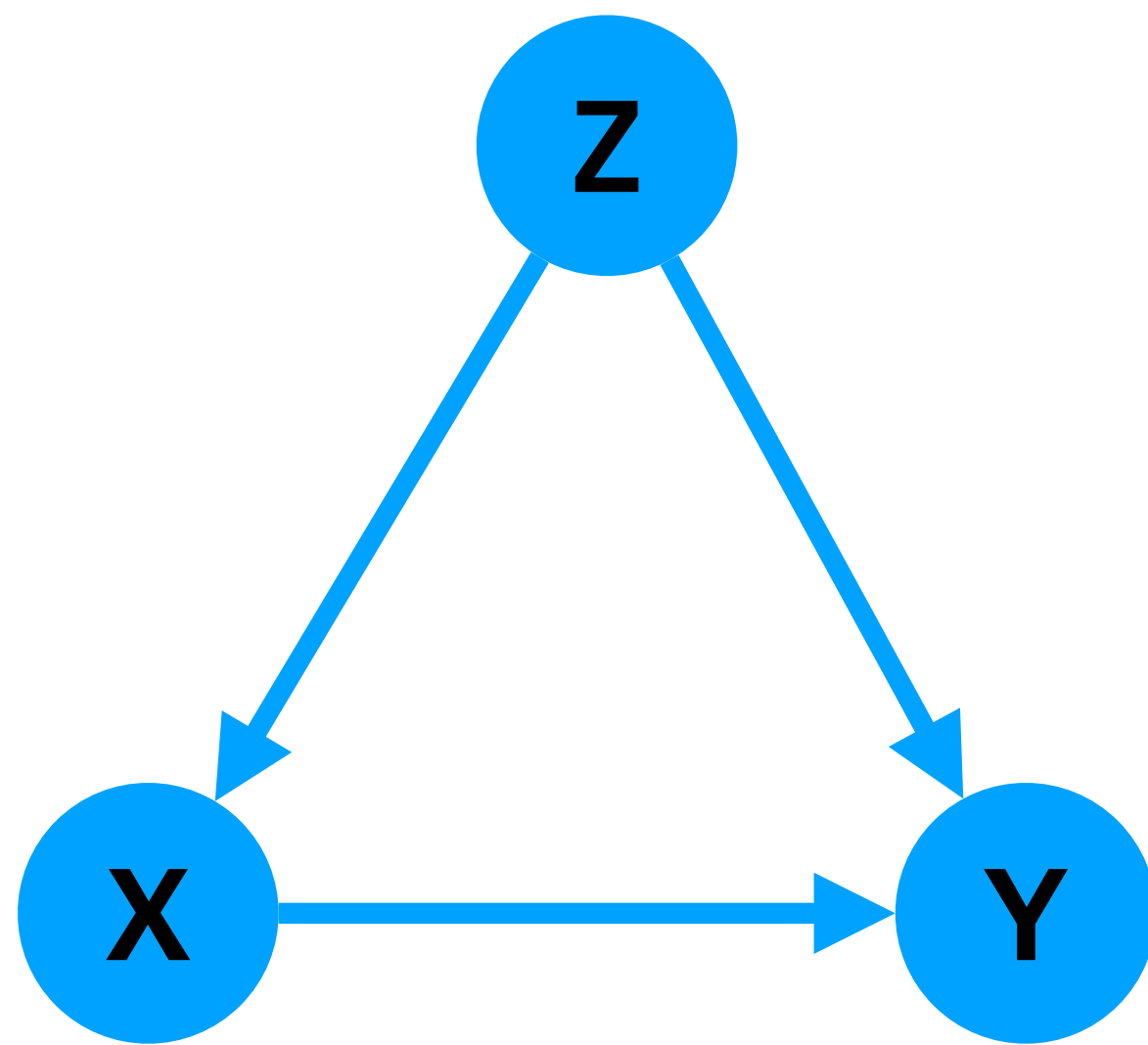


<https://towardsdatascience.com/simpsons-paradox-d2f4d8f08d42>



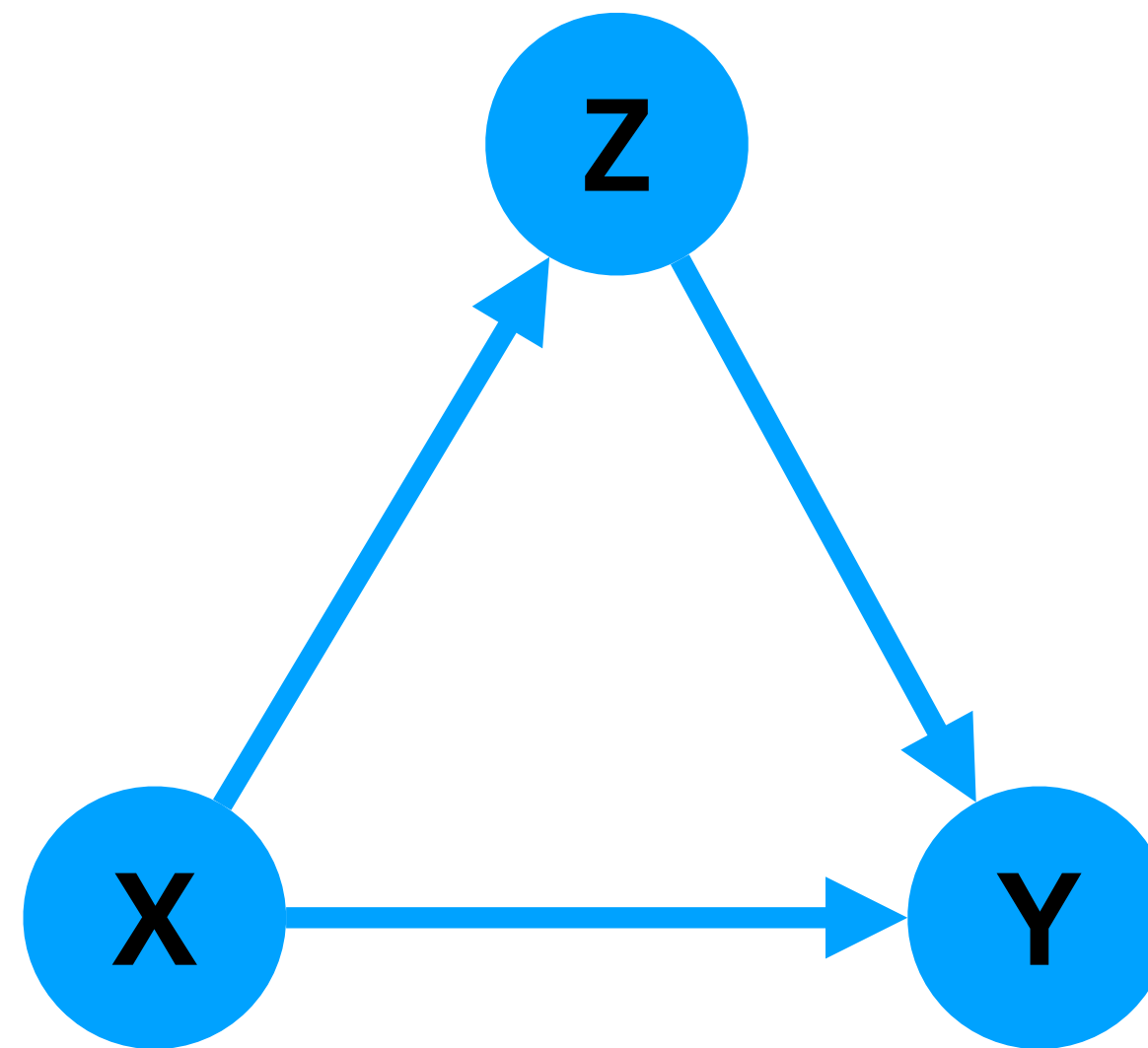
Three types of associations

Confounding



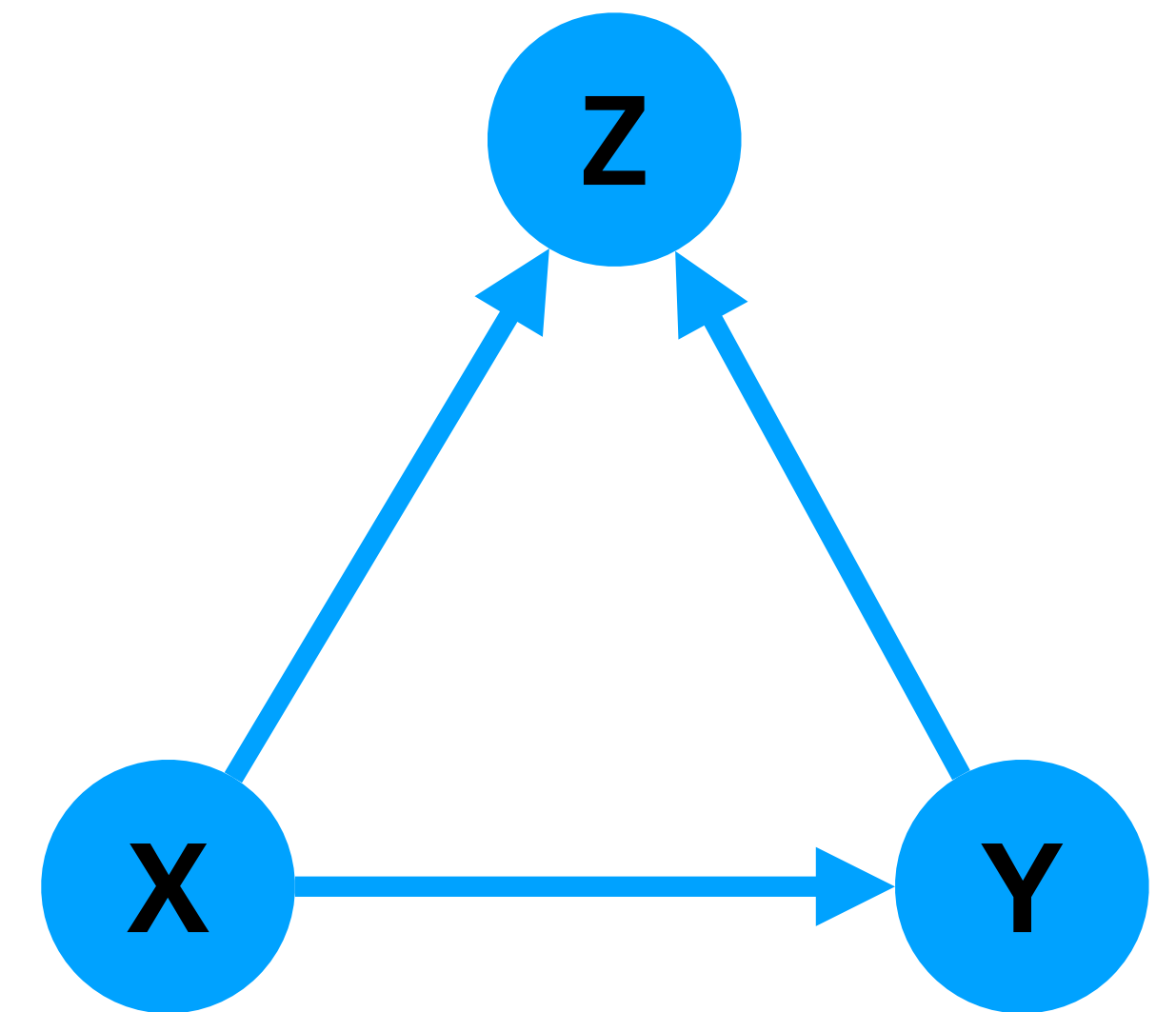
Common cause

Causation



Mediation

Collision



Selection /
endogeneity

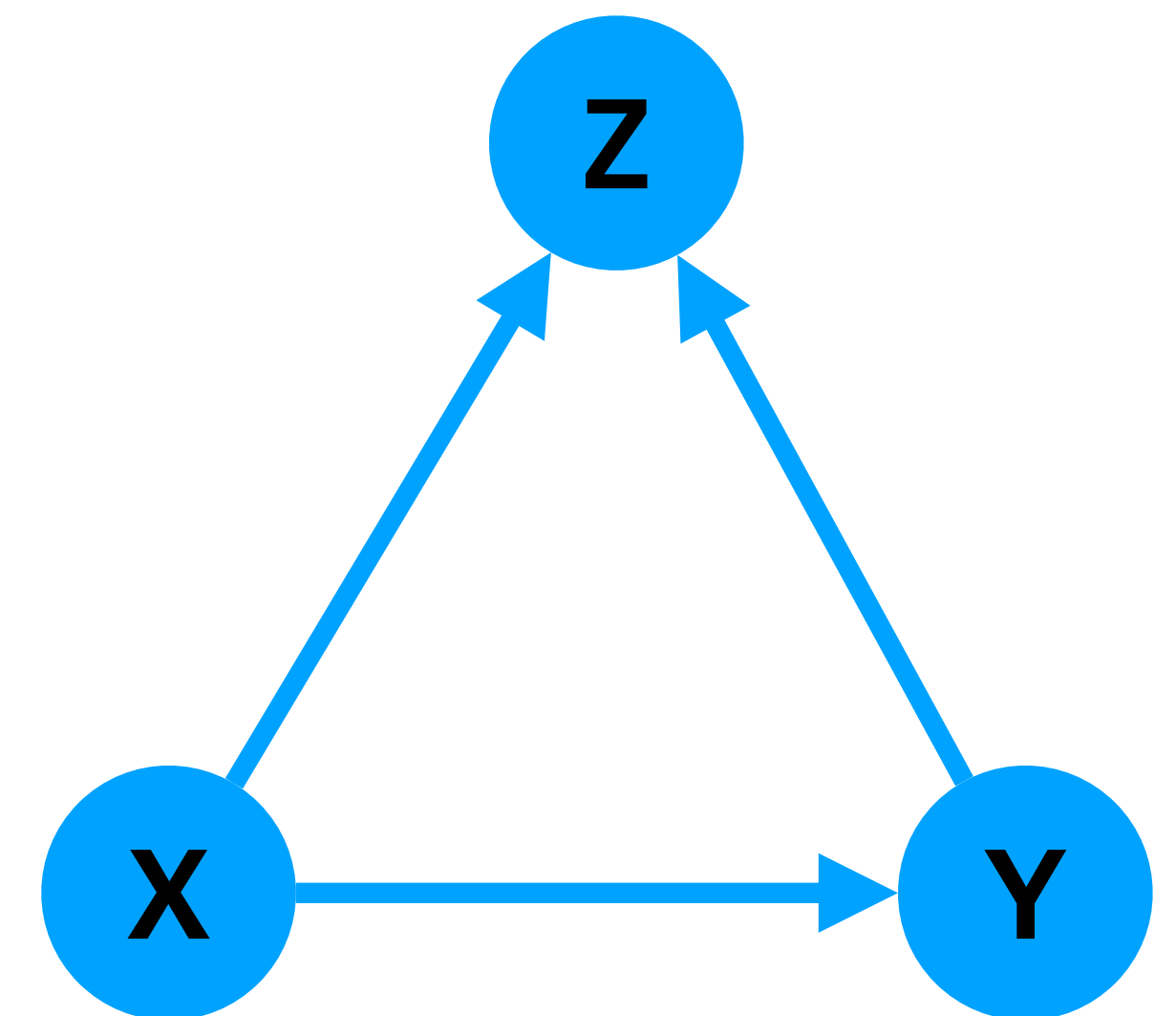
Three types of associations

Endogeneity refers to a situation in statistical modeling where an independent variable is correlated with the error term, meaning there's a missing common cause or a causal relationship between the independent and dependent variables that the model doesn't account for.

This correlation, or **endogeneity**, invalidates the model's causal claims and leads to biased and inconsistent estimates of the relationship between variables.

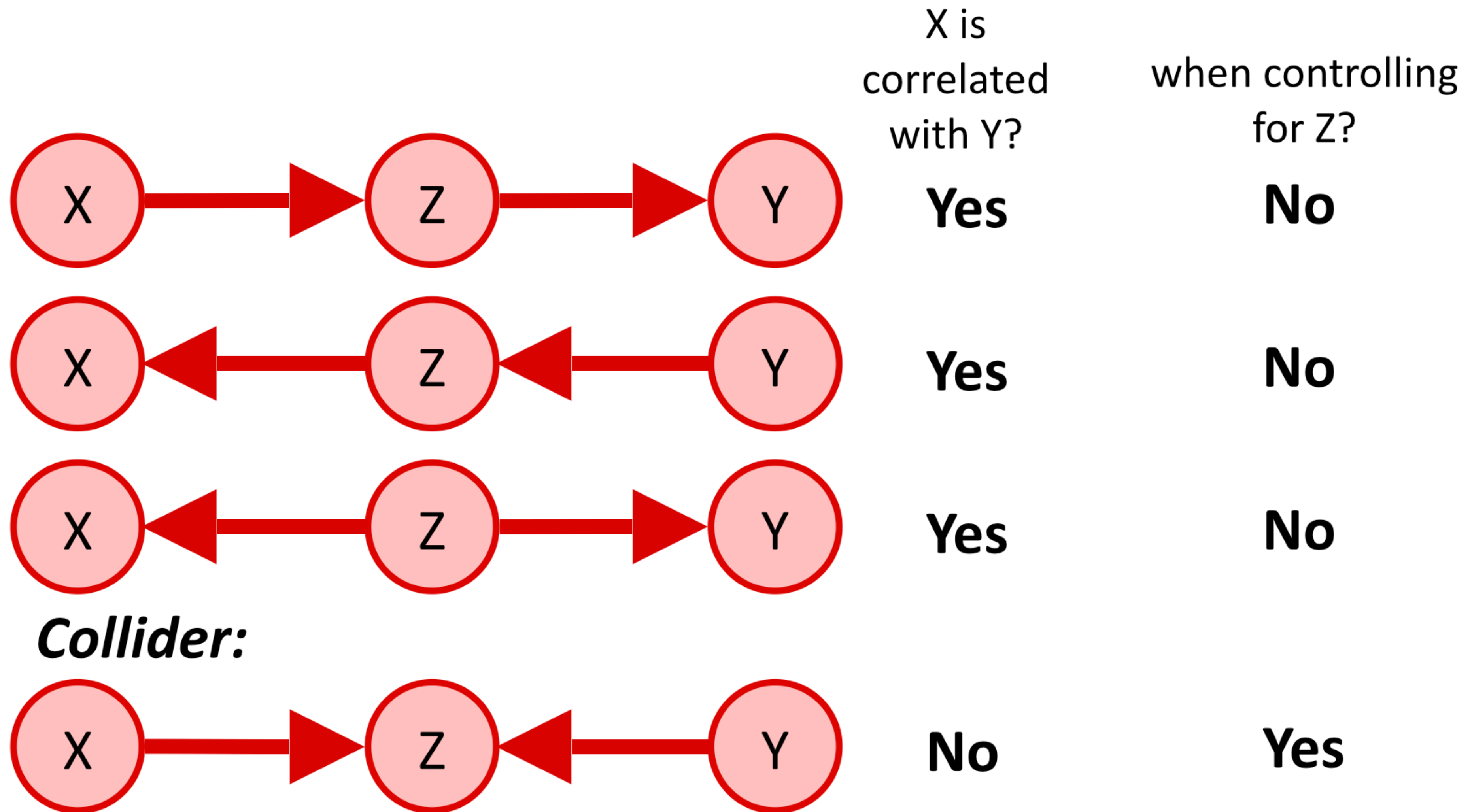
Common causes of **endogeneity** include omitted variable bias, measurement error, and simultaneity, where variables influence each other.

Collision



Selection / endogeneity

Associations



References

Andrew Heiss



Malcolm Barrett



Causal Inference The Mixtape



**Thinking Clearly About
Correlations and Causation:
Graphical Causal Models for
Observational Data**

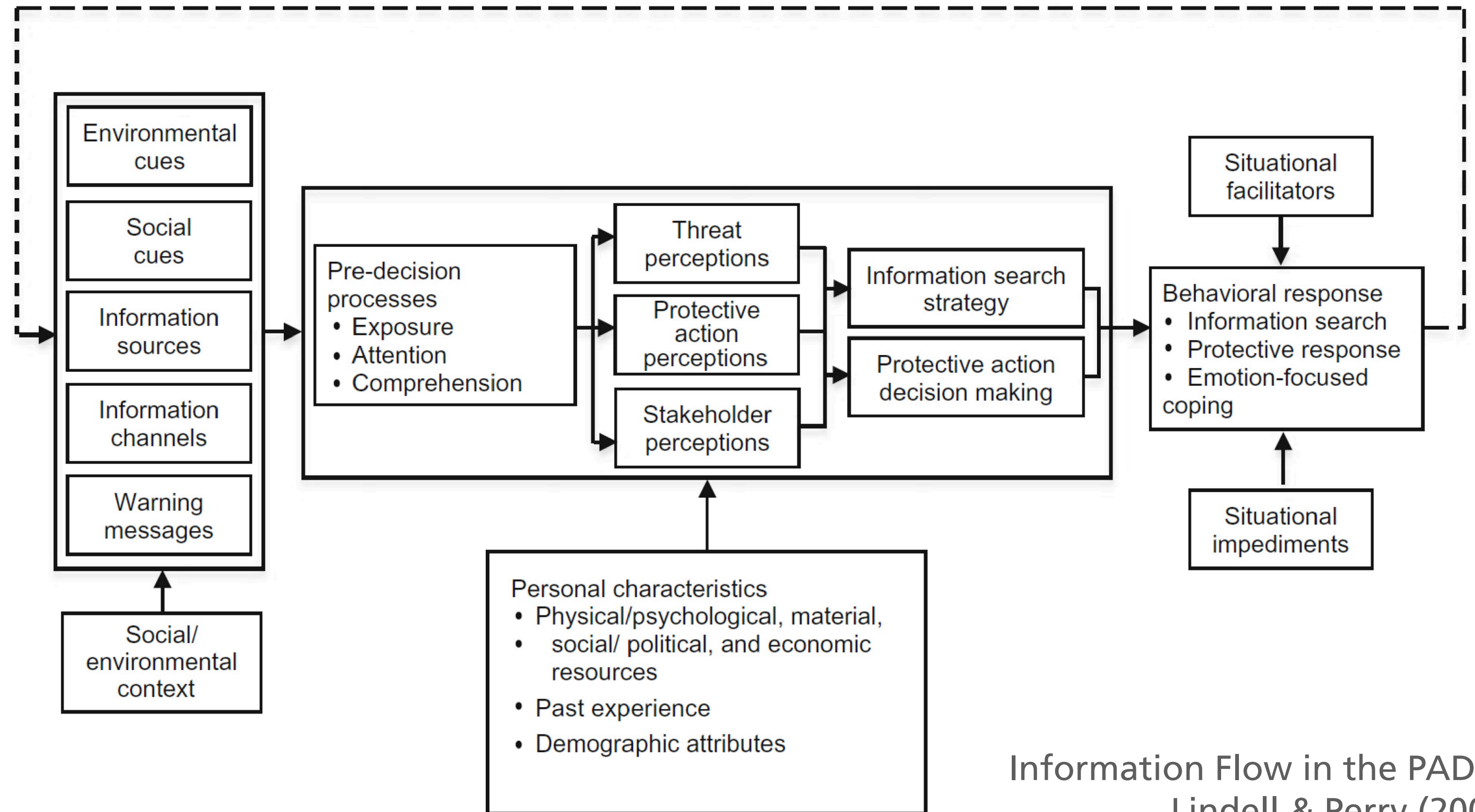


Decision models

Example of application:

Theoretical framework that identifies the critical stages of information processing relevant to household adoption of protective actions

Multistage model



Information Flow in the PADM, Lindell & Perry (2004)

Tripartite Self-Disclosure Decision (TSDD) model

Cognitive, perception, triggers
... MORE NEXT

